

Noosphere Technical Reader — Volume II

A Deeper Apparatus: Mathematics, Mechanism, and the Frontier (2024–2026)

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What this volume is, and what it is not

Volume I was a conceptual map. It introduced the four substrates (LLM, embeddings, NLP extraction, graphs/logic) and gave the deep dives at a level a determined non-specialist could read end-to-end. Volume II assumes Volume I has been read.

This volume does three things Volume I did not. (1) It supplies the *mathematics* that the first reader’s equations only gestured at: linear-algebra derivations, the high-dimensional geometry that explains why embeddings behave strangely, the actual content of information theory, the formal definitions of probabilistic graphical models, and a primer on the differential geometry under “geometric deep learning.” (2) It introduces *frontier material* that postdates or was elided in Volume I: sparse autoencoders done seriously, attribution graphs from Lindsey et al. (2025), influence functions for LLMs, the critique line on contrast-consistent search, function vectors, activation steering, ROME/MEMIT-style knowledge editing, the research programme on Eliciting Latent Knowledge, and the formal apparatus of description logic, AGM belief revision, Dung argumentation, and probabilistic graphical models. (3) It tightens the *methodological* layer: power analysis, proper scoring rules, calibration metrics, Goodhart’s law applied to metric design, the replication crisis, and pre-registration — because the QH-v1 zero-power test in Volume I’s source inventory is exactly the kind of result that demands these tools.

This is not an attempt to be exhaustive. It is an attempt to push the ceiling of useful sophistication up by roughly one academic level: a Volume I reader goes from “I can follow the conceptual story” to “I can read the original papers, evaluate the claims, and design controlled experiments to test them.”

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Part I — The Mathematical Apparatus

1 Linear Algebra You Actually Need

Volume I quoted the attention equation $\text{softmax}(QK^\top/\sqrt{d_k})V$ and the Householder reflection $b' = b - 2(b \cdot \hat{d})\hat{d}$. That suffices for orientation. The following items supply the working machinery a reader will need to evaluate embedding-geometry papers and the contradiction-engine ablations.

1.1 Inner products, norms, and what cosine actually measures

Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. The Euclidean inner product is $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_i a_i b_i = \mathbf{a}^\top \mathbf{b}$. The ℓ_2 norm is $\|\mathbf{a}\|_2 = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$. Cosine similarity is

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2} = \langle \hat{\mathbf{a}}, \hat{\mathbf{b}} \rangle,$$

i.e. the inner product of the ℓ_2 -normalized vectors. Cosine is therefore *purely directional*: it discards magnitude. A retrieval system that uses cosine is implicitly committing to the claim that magnitude does not carry semantic information — a claim that is approximately true for many embedding models trained with normalization, but *not* true for raw attention-head outputs or unnormalized activations.

Three operational consequences for Noosphere. First, if two embedding models differ in their normalization conventions (some normalize at the output, some do not), cosine scores are not portable across them. Second, magnitude sometimes carries confidence or salience signal that cosine throws away; using the unnormalized dot product can recover this but breaks scale invariance. Third, cosine in non-Euclidean inner-product spaces does not measure angle in the visual sense, and several recent papers (notably Park, Choe, Veitch) argue that the correct inner product for concept directions in LLM representations is *not* the Euclidean one but a learned bilinear form.

1.2 Eigendecomposition, SVD, and what counts as a “direction”

For a symmetric real matrix $A \in \mathbb{R}^{n \times n}$, the spectral theorem gives $A = Q\Lambda Q^\top$, where Q is orthogonal and Λ is diagonal. The columns of Q are eigenvectors; the diagonal entries of Λ are eigenvalues. Geometrically, A acts as scaling along orthogonal axes; the axes are the eigenvectors, the scaling factors are the eigenvalues.

For an arbitrary $X \in \mathbb{R}^{m \times n}$, the singular value decomposition is

$$X = U\Sigma V^\top, \quad U \in \mathbb{R}^{m \times m}, V \in \mathbb{R}^{n \times n} \text{ orthogonal}, \Sigma \in \mathbb{R}^{m \times n} \text{ diagonal nonnegative.}$$

The left singular vectors (columns of U) span the column space; the right singular vectors (columns of V) span the row space; the singular values are the lengths of the semi-axes of the image of the unit sphere under X .

SVD matters for embeddings in three distinct ways. (a) Truncated SVD gives the best rank- k approximation in Frobenius norm (Eckart–Young theorem), which is the formal basis for latent-semantic analysis and explains why PCA is informative on embedding matrices. (b) Anisotropy — the spread of singular values — is a diagnostic for whether an embedding space concentrates mass along a few dominant directions, which is exactly what Ethayarajh (2019) showed for BERT and

GPT-2. (c) Sparse autoencoders implicitly recover an overcomplete basis: SVD is the obligatory “baseline” against which any claim that an SAE recovered something novel must be evaluated.

Result (Eckart–Young)

The best rank- k approximation of X in Frobenius norm is $X_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$ where $\sigma_1 \geq \sigma_2 \geq \dots$ are the singular values in decreasing order. Equivalently, $\|X - X_k\|_F^2 = \sum_{i>k} \sigma_i^2$.

1.3 Projection, reflection, and where Householder comes from

Given a unit vector $\mathbf{d} \in \mathbb{R}^n$, the orthogonal projection of $\mathbf{b}$ onto the line $\text{span}(\mathbf{d})$ is $(\mathbf{b} \cdot \mathbf{d})\mathbf{d}$. The projection onto the hyperplane $\mathbf{d}^\perp$ is therefore $\mathbf{b} - (\mathbf{b} \cdot \mathbf{d})\mathbf{d}$. A *reflection* across that hyperplane keeps the perpendicular component and reverses the parallel component:

$$H_{\mathbf{d}} \mathbf{b} = \mathbf{b} - 2(\mathbf{b} \cdot \mathbf{d})\mathbf{d} = (I - 2\mathbf{d}\mathbf{d}^\top)\mathbf{b}.$$

The Householder matrix $H_{\mathbf{d}} = I - 2\mathbf{d}\mathbf{d}^\top$ is symmetric ($H = H^\top$), orthogonal ($HH^\top = I$), and involutory ($H^2 = I$). Its determinant is -1 . It is the simplest example of a non-trivial isometry of $\mathbb{R}^n$, and the natural building block for QR factorization in numerical linear algebra (Trefethen–Bau, Lecture 10).

Noosphere uses $H_{\hat{\mathbf{d}}}$ to reflect an embedding across an estimated contradiction hyperplane. The mathematics says: if the contradiction direction is correctly estimated and contradiction really is the antipodal pair across that hyperplane, then $H_{\hat{\mathbf{d}}} \mathbf{b} - \mathbf{a}$ should concentrate mass along the contradiction direction. The Hoyer sparsity score is one operationalization of “concentrates mass.” Two failure modes follow immediately. If $\hat{\mathbf{d}}$ is poorly estimated, the reflection rotates the embedding into nonsense space. If contradiction is not a single direction but a context-dependent subspace, a fixed $\hat{\mathbf{d}}$ projects away precisely the information that distinguishes one contradiction type from another.

1.4 Positive semidefinite matrices, kernels, and Mahalanobis distance

A symmetric matrix A is positive semidefinite (PSD) iff $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$. PSD matrices are the matrices that arise as covariance matrices, Gram matrices, and kernels. Any PSD matrix admits a Cholesky factorization $A = LL^\top$ with L lower-triangular.

The Mahalanobis distance with respect to a PSD matrix Σ is $\sqrt{(\mathbf{x} - \mathbf{y})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{y})}$. It is the natural distance when coordinates are correlated and have unequal variances. For a Noosphere geometry that uses raw cosine on embeddings whose coordinates have wildly unequal variances, the Mahalanobis distance under the empirical covariance is the obvious better-calibrated alternative. “Whitening” an embedding space is precisely the change of basis that turns Mahalanobis distance into Euclidean distance — multiply each vector by $\Sigma^{-1/2}$, after which coordinates are decorrelated and unit-variance.

1.5 Gradients, Jacobians, and the matrix-calculus you need

Almost all of the relevant calculus is captured by three identities. Let $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$. Then $\nabla_{\mathbf{x}} f = \mathbf{a}$. Let $g(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$ with A symmetric. Then $\nabla_{\mathbf{x}} g = 2A\mathbf{x}$. Let $h(W) = \|W\mathbf{x} - \mathbf{y}\|_2^2$. Then $\nabla_W h = 2(W\mathbf{x} - \mathbf{y})\mathbf{x}^\top$. From these three, plus the chain rule, you can derive backpropagation in any transformer.

1.6 Recommended reading for Part I.1

- Lloyd N. Trefethen & David Bau, *Numerical Linear Algebra*, SIAM 1997 — Lectures 4 (the SVD), 5 (more on SVD), 10 (Householder triangularization), 11 (least squares). Paid but essential.
- Stephen Boyd & Lieven Vandenberghe, *Introduction to Applied Linear Algebra*, Cambridge 2018; free at <https://web.stanford.edu/~boyd/vmls/>.
- Kaare Brandt Petersen & Michael Syskind Pedersen, *The Matrix Cookbook*; free PDF, the standard quick reference for matrix derivatives.
- Roger A. Horn & Charles R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge 2013 — the definitive reference if you need a theorem about matrices in any form. Paid.

2 High-Dimensional Geometry: Why Intuition Lies

The single most important fact about embeddings is that they live in spaces where two- and three-dimensional intuition is systematically wrong. A reader who has not internalized this fact will, sooner or later, declare a result real that is actually an artifact of dimensionality.

2.1 Concentration of measure

Consider $\mathbf{X} \sim \mathcal{N}(0, I_n)$. Then $\|\mathbf{X}\|_2^2$ has a chi-squared distribution with n degrees of freedom, mean n and variance $2n$. By Chebyshev or by sharper concentration inequalities, with high probability,

$$\|\mathbf{X}\|_2 \in [\sqrt{n} - O(1), \sqrt{n} + O(1)].$$

All the mass of a high-dimensional standard Gaussian sits in a thin shell of radius $\sqrt{n}$. The intuition of a “ball” filling space breaks immediately: for $n = 768$ (BERT-base) or $n = 1536$ (OpenAI ada-002), nearly all random points are at almost exactly the same distance from the origin.

A related consequence: two independent random vectors in $\mathbb{R}^n$ are, with high probability, nearly orthogonal. The expected cosine between two random Gaussian unit vectors is 0 with variance $1/n$. For $n = 1536$, this means random pairs have $\cos \sim \mathcal{N}(0, 1/1536)$, so a cosine of 0.05 is already several standard deviations away from random. The practical implication for Noosphere is that cosine thresholds calibrated by intuition (“> 0.7 is similar”) are nearly useless without knowledge of the embedding model’s empirical cosine distribution on *negative* pairs.

2.2 The Johnson–Lindenstrauss lemma

Result (Johnson–Lindenstrauss, 1984)

For any $0 < \epsilon < 1$, any set S of m points in $\mathbb{R}^n$, and any $k \geq \Omega(\epsilon^{-2} \log m)$, there is a linear map $f: \mathbb{R}^n \rightarrow \mathbb{R}^k$ such that for all $\mathbf{u}, \mathbf{v} \in S$,

$$(1 - \epsilon) \|\mathbf{u} - \mathbf{v}\|_2^2 \leq \|f(\mathbf{u}) - f(\mathbf{v})\|_2^2 \leq (1 + \epsilon) \|\mathbf{u} - \mathbf{v}\|_2^2.$$

A random orthogonal projection works with high probability.

J–L says high-dimensional pairwise distances are preserved by random low-rank projections. This is why random projection works for nearest-neighbor search and why approximate methods like FAISS or HNSW can return near-exact results without storing full embeddings. For Noosphere, the

J–L lemma is the quantitative argument that 1536-dimensional OpenAI embeddings carry roughly $O(\log m)$ “bits of distance information” per query, where m is the corpus size. If the firm’s corpus has 10^5 claims, that is ~ 50 bits per distance — ample for retrieval, marginal for fine-grained logical discrimination.

2.3 Anisotropy, the cone effect, and the hubness problem

Three pathological phenomena recur in contextual embeddings; all three threaten naive Noosphere geometry.

Anisotropy. Ethayarajh (2019, “How Contextual are Contextualized Word Representations?”) showed empirically that BERT, GPT-2, and ELMo representations occupy a narrow cone rather than the full sphere. Pairwise cosine between random tokens in BERT layer 11 averages above 0.6. Without correction, any cosine-based metric will systematically over-report similarity.

The cone effect. Mu & Viswanath (2018, “All-but-the-Top”) gave a post-hoc correction: subtract the mean, then remove the top few principal components. The result is sharply more isotropic. “Whitening” as defined above is the principled version.

Hubness. In high dimensions, some points become nearest neighbors to *disproportionately many* other points. These “hubs” break the assumption that nearest-neighbor relations are symmetric and balanced. For a graph built from nearest-neighbor relations on raw embeddings, hubs become artificial central nodes whose centrality reflects geometry, not semantic importance.

2.4 What the Noosphere geometry pipeline must do

A geometry-based contradiction or coherence detector that does not (a) measure and report anisotropy of its embedding space, (b) compare against an isotropic-by-construction baseline (e.g. random projection or whitened embeddings), and (c) calibrate distance thresholds on actual negative-pair distributions cannot, in principle, distinguish its claims from these artifacts. The QH-v1 result that the firm probe underperformed cosine is *exactly* the kind of warning that high-dimensional artifacts may be contaminating the signal. The remedy is not to abandon the hypothesis but to add the controls.

2.5 Recommended reading for Part I.2

- Roman Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, Cambridge 2018. Free draft at <https://www.math.uci.edu/~rvershyn/papers/HDP-book/HDP-book.pdf>. The standard reference; chapters 3 and 5 cover concentration and J–L.
- Avrim Blum, John Hopcroft, Ravindran Kannan, *Foundations of Data Science*, Cambridge 2020. Free draft online. Chapter 2 is the cleanest short introduction to high-dimensional geometry for non-mathematicians.
- Kawin Ethayarajh, *How Contextual are Contextualized Word Representations?*, EMNLP 2019. arXiv:1909.00512. <https://arxiv.org/abs/1909.00512>.
- Jiaqi Mu & Pramod Viswanath, *All-but-the-Top: Simple and Effective Postprocessing for Word Representations*, ICLR 2018. arXiv:1702.01417. <https://arxiv.org/abs/1702.01417>.
- Miloš Radovanović, Alexandros Nanopoulos, Mirjana Ivanović, *Hubs in Space: Popular Nearest Neighbors in High-Dimensional Data*, JMLR 2010.

3 Probability, Calibration, and Proper Scoring Rules

The QH-v1 source inventory documents a “zero-power label test under the frozen threshold.” That phrase has a precise meaning in statistics, and every word matters. This subsection supplies the apparatus.

3.1 What “power” means

For a hypothesis test of H_0 against H_1 , the type-I error rate α is the probability of rejecting H_0 when it is true; the type-II error rate β is the probability of failing to reject H_0 when H_1 is true. The *power* of the test is $1 - \beta$: the probability of correctly rejecting H_0 when H_1 holds. Power depends on α , sample size, and effect size.

A “zero-power” result means $\beta \approx 1$: even if the alternative were true, the test would not detect it. The most common cause is insufficient sample size relative to the effect size you are trying to detect. A second cause, which is what happened in QH-v1’s ablation, is that the test statistic is *degenerate* — the test always reports the same label, so it cannot distinguish anything. The correct response to a zero-power test is not to reinterpret the result but to redesign the test until it has power against the alternative you care about.

3.2 Power analysis

To plan a study with adequate power, you specify (i) α (conventionally 0.05), (ii) desired power (conventionally 0.8 or 0.9), (iii) the smallest effect size you want to detect, and (iv) the variance structure. Then you solve for n . For comparing two proportions (p_1 vs. p_2) with balanced groups, the approximate sample size per group is

$$n \approx \frac{(z_{1-\alpha/2}\sqrt{2\bar{p}(1-\bar{p})} + z_{1-\beta}\sqrt{p_1(1-p_1) + p_2(1-p_2)})^2}{(p_1 - p_2)^2},$$

where $\bar{p} = (p_1 + p_2)/2$ and z_q is the standard normal quantile.

Example 3.1. To detect a contradiction-detector accuracy of $p_1 = 0.75$ vs. a baseline of $p_2 = 0.70$ with $\alpha = 0.05$ and power 0.80, $n \approx 1100$ per group. QH-v1’s reported sample size, if smaller than this, was structurally underpowered to distinguish a five-percentage-point lift.

3.3 Proper scoring rules

A scoring rule assigns a score to a probabilistic forecast $\mathbf{p}$ given an outcome y . A scoring rule is *proper* if reporting one’s true belief maximizes expected score; it is *strictly proper* if the truth is the unique maximizer. The two scoring rules to know:

The *Brier score* for binary outcomes is $BS = (p - y)^2$, where $p \in [0, 1]$ is the forecast and $y \in \{0, 1\}$ is the realization. It is strictly proper.

The *logarithmic score* is $LS = -\log p_y$, where p_y is the forecast probability of the realized outcome. It is also strictly proper, but diverges if you assign probability zero to an outcome that occurs — forcing calibration of tail beliefs. It is the natural information-theoretic score: $\mathbb{E}_y[-\log p_y]$ is cross-entropy.

Why this matters for Noosphere. Any coherence score that the firm treats as a probability of truth must, eventually, be evaluated by a proper scoring rule against realized outcomes. A score

that is high for false claims and low for true claims fails as a probability, even if it sounds rigorous as a number.

3.4 Calibration

A binary forecaster is *calibrated* if, among events to which it assigned probability p , the empirical frequency of occurrence is p . The *expected calibration error* (ECE) bins the predictions and reports

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|,$$

where B_b is the b -th bin, $\text{acc}(B_b)$ is empirical accuracy in the bin, and $\text{conf}(B_b)$ is mean predicted probability. A *reliability diagram* plots accuracy vs. confidence per bin; a perfectly calibrated forecaster is on the diagonal.

Two practical post-hoc calibration methods. *Platt scaling* fits a logistic regression $\hat{p} = \sigma(a \cdot s + b)$ from raw score s to calibrated probability $\hat{p}$, on a held-out calibration set. *Isotonic regression* is non-parametric: it fits a monotonic step function, which handles non-sigmoid miscalibration better but overfits on small samples.

Operational rule for Noosphere. No coherence or contradiction *score* should be exposed in a decision trace as a probability without first being calibrated against held-out labels with a reported ECE and a reliability diagram. A score reported as “0.82 confidence” that is empirically right 60% of the time is not a confidence; it is a number.

3.5 Bayesian networks formally: d-separation

The Bayesian Belief Layer in Volume I deserves the actual graphical model definition. A Bayesian network is a pair (G, P) where $G = (V, E)$ is a DAG over random variables V and P is a joint distribution that factorizes as $P(V) = \prod_i P(V_i | \text{Pa}(V_i))$.

The key inferential tool is *d-separation*. A path is *blocked* by a set Z if it contains either (a) a chain $A \rightarrow B \rightarrow C$ or a fork $A \leftarrow B \rightarrow C$ where $B \in Z$, or (b) a collider $A \rightarrow B \leftarrow C$ where neither B nor any descendant of B is in Z . Variables X and Y are d-separated given Z if every path between them is blocked by Z . D-separation implies conditional independence in the joint distribution.

Why this matters for Noosphere. The cascade graph contains many forks and colliders. Naive marginalization can introduce spurious correlations through unblocked collider paths. The firm’s policy of deterministically dropping cycle-closing edges, while sensible, must be accompanied by an explicit accounting of which conditional independences hold in the projected DAG — otherwise the Bayesian update will be reading correlations as evidence when they are graph artifacts.

3.6 Recommended reading for Part I.3

- Joseph K. Blitzstein & Jessica Hwang, *Introduction to Probability*, 2nd ed., 2019. Free draft and Harvard Stat 110 lectures online. The pedagogical gold standard.
- Andrew Gelman, John Carlin, Hal Stern, David Dunson, Aki Vehtari, Donald Rubin, *Bayesian Data Analysis*, 3rd ed., CRC 2013. Free draft at <http://www.stat.columbia.edu/~gelman/book/>. The standard reference; Chapter 6 (model checking) is the closest match to Noosphere’s needs.

- Daphne Koller & Nir Friedman, *Probabilistic Graphical Models: Principles and Techniques*, MIT Press 2009. The definitive textbook on Bayesian networks, Markov random fields, and inference. Paid; this is the serious reference.
- Tilmann Gneiting & Adrian Raftery, *Strictly Proper Scoring Rules, Prediction, and Estimation*, JASA 2007. Free preprint widely circulated. The reference paper on proper scoring rules.
- Chuan Guo, Geoff Pleiss, Yu Sun, Kilian Weinberger, *On Calibration of Modern Neural Networks*, ICML 2017. arXiv:1706.04599. <https://arxiv.org/abs/1706.04599>. Showed that modern deep networks are *worse*-calibrated than older shallow networks — a result every LLM-using firm should know.
- Volodymyr Kuleshov, Nathan Fenner, Stefano Ermon, *Accurate Uncertainties for Deep Learning Using Calibrated Regression*, ICML 2018.

4 Information Theory

Noosphere’s six-layer coherence engine includes an “information-theoretic compression” layer. Without a working definition of information, this is a formula in search of a meaning. The following is the minimum.

4.1 Entropy, cross-entropy, KL

For a discrete distribution P over a finite alphabet $\mathcal{X}$, the Shannon entropy is

$$H(P) = - \sum_{x \in \mathcal{X}} P(x) \log P(x).$$

H is the expected number of bits (with $\log_2$) needed to encode samples from P using an optimal code. The cross-entropy of Q relative to P is $H(P, Q) = - \sum_x P(x) \log Q(x)$. The Kullback–Leibler divergence is

$$D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} = H(P, Q) - H(P).$$

$D_{\text{KL}}(P \parallel Q) \geq 0$ with equality iff $P = Q$ a.e.; it is asymmetric and not a metric. Loosely: $D_{\text{KL}}(P \parallel Q)$ is the expected excess bits paid by coding samples from P with a code optimized for Q .

Implication for Noosphere. Cross-entropy is the natural loss for classification; it is what an LLM minimizes during pre-training. A “compression”-style coherence layer that scores a set of claims by their description length under a reference model is performing an estimate of $H(\text{claims} \mid \text{model})$. Two claim sets are then “equally coherent” if the model can compress them equally well — which conflates *semantic* coherence with *conformity to the reference model*. Compression-based coherence under a sycophantic LLM will reward sycophancy.

4.2 Mutual information

For random variables X, Y ,

$$I(X; Y) = D_{\text{KL}}(P_{XY} \parallel P_X \otimes P_Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X).$$

$I \geq 0$ with equality iff X and Y are independent. Mutual information is the right scalar for “how much does X tell me about Y .”

For Noosphere, a candidate coherence measure between two claim embeddings X, Y is their mutual information — not their cosine. Estimating I in high dimensions is notoriously hard (the MINE estimator and CLUB upper bound are state of the art), but the conceptual move is sound: cosine is a zeroth-order statistic; mutual information is the entire dependence structure.

4.3 Fisher information and the natural gradient

For a parametric family $\{P_\theta\}$, the Fisher information matrix is $I(\theta) = \mathbb{E}_\theta[\nabla_\theta \log P_\theta \cdot \nabla_\theta \log P_\theta^\top]$. It is a Riemannian metric on the statistical manifold $\{P_\theta\}$; the *natural gradient* $I(\theta)^{-1} \nabla_\theta \mathcal{L}$ is the gradient direction in which a step of fixed KL size produces the largest decrease in loss. This is the geometric content of the Park–Choe–Veitch “causal inner product” for LLM representations.

4.4 Recommended reading for Part I.4

- Thomas M. Cover & Joy A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley 2006. The standard textbook. Paid.
- David J.C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge 2003. Free PDF at <https://www.inference.org.uk/itila/book.html>. The free alternative; read instead of, or alongside, Cover & Thomas. Chapters 1–6 and 22–24.
- Shun-ichi Amari, *Information Geometry and Its Applications*, Springer 2016. Paid. The reference for treating $\{P_\theta\}$ as a manifold with the Fisher information metric.
- Mohamed Ishmael Belghazi et al., *MINE: Mutual Information Neural Estimation*, ICML 2018. arXiv:1801.04062. The starting point for high-dimensional MI estimation.

5 Optimization and the Loss Landscape

5.1 Gradient descent, stochastic gradient descent, Adam

Plain gradient descent updates $\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}(\theta_t)$. SGD replaces the full-data gradient with a stochastic estimate from a mini-batch. The implicit regularization properties of SGD (noise that helps escape sharp minima, bias toward “flat” minima associated with better generalization) are a serious area of theoretical research.

Adam adapts learning rates per parameter using first and second moment estimates:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) \nabla \mathcal{L}_t, \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) (\nabla \mathcal{L}_t)^{\odot 2}, \\ \hat{m}_t &= m_t / (1 - \beta_1^t), \quad \hat{v}_t = v_t / (1 - \beta_2^t), \\ \theta_{t+1} &= \theta_t - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon). \end{aligned}$$

The bias correction in $\hat{m}, \hat{v}$ is necessary because at $t = 1$ the moment estimates are themselves biased toward zero. AdamW (Loshchilov & Hutter, 2017) decouples weight decay from the gradient step, fixing a long-standing bug in Adam’s interaction with ℓ_2 regularization. AdamW is now the dominant optimizer for transformer training.

5.2 Why the loss landscape matters

Modern deep networks are massively overparameterized; the loss landscape has exponentially many local minima, most of which are nearly equivalent in training loss but differ substantially in generalization. “Flat” minima (small Hessian eigenvalues) generalize better than “sharp” minima (Keskar et al., 2017). This is one mechanistic ingredient in the empirical observation that bigger models trained longer with more data generalize better than the bias-variance trade-off would predict.

5.3 Recommended reading for Part I.5

- Stephen Boyd & Lieven Vandenberghe, *Convex Optimization*, Cambridge 2004. Free PDF at <https://web.stanford.edu/~boyd/cvxbook/>.
- Sébastien Bubeck, *Convex Optimization: Algorithms and Complexity*, Foundations and Trends in ML, 2015. arXiv:1405.4980. Free. The reference for non-trivial convergence-rate analyses.
- Diederik P. Kingma & Jimmy Ba, *Adam: A Method for Stochastic Optimization*, ICLR 2015. arXiv:1412.6980.
- Ilya Loshchilov & Frank Hutter, *Decoupled Weight Decay Regularization* (AdamW), ICLR 2019. arXiv:1711.05101.
- Nitish Shirish Keskar et al., *On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima*, ICLR 2017. arXiv:1609.04836.

6 Differential Geometry, Just Enough

The Bronstein et al. monograph and several recent representation-engineering papers (Park, Choe, Veitch in particular) use the language of differential geometry: manifold, tangent space, geodesic, Lie group. A reader who skipped this in undergraduate math will get more out of those papers after the following minimal primer.

6.1 Manifolds, tangent spaces, geodesics

A smooth n -manifold M is a topological space locally homeomorphic to $\mathbb{R}^n$ with smooth transition maps between charts. Examples: $\mathbb{R}^n$ itself, the n -sphere S^n , the orthogonal group $O(n)$, the manifold of k -rank matrices.

The *tangent space* $T_p M$ at a point $p \in M$ is the n -dimensional vector space of velocity vectors of smooth curves through p . A Riemannian metric g assigns to each $T_p M$ an inner product $g_p(\cdot, \cdot)$ varying smoothly with p . A *geodesic* is a curve of locally minimum length under g .

Why this matters for representations. The Linear Representation Hypothesis effectively claims that concepts are tangent vectors at a “neutral” point in representation space, and that translation along those vectors corresponds to concept variation. The Park–Choe–Veitch “causal inner product” is a specific choice of Riemannian metric on representation space. The manifold hypothesis says embedded data lives on a low-dimensional submanifold; “coherence as geometry” is then a claim about the curvature or connectivity properties of that submanifold.

6.2 Recommended reading for Part I.6

- John M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer 2013. Paid but the standard. Chapters 1–3 are the minimum.
- John M. Lee, *Introduction to Riemannian Manifolds*, Springer 2018. Chapter 1–2 give what you need.
- Loring W. Tu, *An Introduction to Manifolds*, 2nd ed., Springer 2011. Friendlier than Lee.
- Sigurdur Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, AMS 2001. The reference for symmetry-based machine learning; advanced.

Part II — LLM Internals: Mechanism Beyond Architecture

7 Tokenization in Detail

Volume I noted that “tokenization is already a modeling choice.” Three algorithms dominate production LLMs and each has consequences for Noosphere.

7.1 BPE, WordPiece, SentencePiece

Byte-Pair Encoding (Sennrich et al., 2016, originally Gage 1994 for compression) builds a vocabulary by iteratively merging the most frequent adjacent byte pair in a corpus. The result is a vocabulary of subword units sized to fit a target budget (typically 30k–100k). *Byte-level BPE*, used by GPT-2 onward, operates on raw bytes rather than characters, which eliminates the need for language-specific normalization and handles any Unicode string.

WordPiece (Schuster & Nakajima, 2012; used by BERT) merges by likelihood rather than frequency: pick the merge that maximizes the likelihood of the corpus under a unigram language model over subwords.

SentencePiece (Kudo & Richardson, 2018) is a unigram-language-model tokenizer that operates directly on raw text streams without pretokenization (no whitespace assumptions). Now the dominant choice for non-English and multilingual models including Claude, Llama, Mistral.

7.2 Why this matters for Noosphere

Three consequences. (i) Token-aligned offsets returned by an extractor may not match human sentence boundaries; if evidence pointers store byte offsets from a sentence-splitter and the model returns offsets from its tokenizer, they will not always agree. (ii) Domain-specific vocabulary (technical terms, proper nouns, neologisms) is frequently fragmented into many sub-tokens. Embeddings of multi-token spans are pooled from constituent token representations and inherit pooling artifacts; the choice of pooling (mean, max, last-token, CLS) materially changes the resulting vector. (iii) Cross-model embedding comparisons are not portable across tokenizers — the OpenAI embedding of “Quintin Hypothesis” and the Voyage embedding of the same string are not in the same space, and any cosine comparison between them is meaningless.

7.3 Recommended reading

- Rico Sennrich, Barry Haddow, Alexandra Birch, *Neural Machine Translation of Rare Words with Subword Units*, ACL 2016. arXiv:1508.07909. <https://arxiv.org/abs/1508.07909>.
- Taku Kudo & John Richardson, *SentencePiece: A Simple and Language Independent Subword Tokenizer*, EMNLP 2018. arXiv:1808.06226.
- Sander Land, *Fishing for Magikarp: Automatically Detecting Under-trained Tokens in Large Language Models*, EMNLP 2024. arXiv:2405.05417. A startling result: every production LLM contains tokens whose embeddings are essentially random because they appeared too rarely in pre-training. For any Noosphere extractor handling jargon, this is a real failure mode.

8 Attention, Derived and Decomposed

8.1 Why the scaling factor $1/\sqrt{d_k}$

Let $\mathbf{q}, \mathbf{k} \in \mathbb{R}^{d_k}$ have entries i.i.d. with mean 0 and variance 1. Then $\langle \mathbf{q}, \mathbf{k} \rangle$ has mean 0 and variance d_k . For large d_k , dot products thus have magnitudes proportional to $\sqrt{d_k}$, which pushes softmax into saturated regions where gradients vanish. Dividing by $\sqrt{d_k}$ restores unit-variance dot products and well-conditioned gradients. This is the entire justification for the $1/\sqrt{d_k}$ in Vaswani et al.

8.2 Softmax: numerical stability and temperature

The softmax function $\text{softmax}(\mathbf{z})_i = e^{z_i} / \sum_j e^{z_j}$ is numerically unstable for large z_i . The standard trick is to subtract $z_{\max} = \max_j z_j$ before exponentiating; the output is unchanged but no exponent overflows.

A temperature parameter τ enters as $\text{softmax}(\mathbf{z}/\tau)$. As $\tau \rightarrow 0$, softmax becomes argmax (deterministic); as $\tau \rightarrow \infty$, it becomes uniform. In attention, temperature scales the sharpness of the attention distribution; in sampling, it scales the sharpness of the next-token distribution.

8.3 Attention heads as algorithms

The Elhage et al. “Mathematical Framework for Transformer Circuits” (2021) made the following decomposition canonical. A single attention head consists of two products: $W_{QK} = W_Q W_K^\top \in \mathbb{R}^{d \times d}$ (the *QK circuit*) and $W_{OV} = W_O W_V \in \mathbb{R}^{d \times d}$ (the *OV circuit*). The QK circuit determines *which* tokens attend to which; the OV circuit determines *what* is copied from attended-to tokens into the destination. Treating these as two independent learned matrices is a powerful interpretive move: it lets you ask, of any head, “does its QK circuit implement a recognizable selection rule (e.g. previous token, matching token, syntactic head)?” and separately “does its OV circuit copy a recognizable feature?”

Induction heads (Olsson et al., 2022) are a specific attention-head pattern with QK matching previous occurrences of the current token and OV copying the token that followed the previous occurrence. They implement a learnable form of in-context pattern completion and are empirically responsible for a large fraction of in-context learning in transformers.

8.4 Positional encodings: sinusoidal, learned, RoPE, ALiBi

The original transformer used sinusoidal positional encodings: $\text{PE}_{pos,2i} = \sin(pos/10000^{2i/d})$, $\text{PE}_{pos,2i+1} = \cos(pos/10000^{2i/d})$. These are added to token embeddings; they have no learnable parameters.

Learned absolute positions (BERT, GPT-2): a position-indexed embedding table is learned. Problem: cannot extrapolate beyond training length.

Rotary Position Embedding (RoPE, Su et al., 2021, arXiv:2104.09864) applies a rotation to query and key vectors as a function of position. The relative angle between $\mathbf{q}_i$ and $\mathbf{k}_j$ depends only on $i - j$, so attention scores naturally become functions of relative position. RoPE is now standard in Llama, Mistral, and most open-source LLMs.

ALiBi (Press, Smith, Lewis, 2021, arXiv:2108.12409) skips positional embeddings entirely and adds a linear bias to attention scores proportional to the distance between tokens. Extrapolates remarkably well to sequences longer than seen in training.

8.5 In-context learning: three theories

ICL is the phenomenon that a sufficiently capable LLM solves new tasks given only examples in its prompt, without weight updates. Three explanatory frameworks:

Implicit Bayesian inference. Xie, Raghunathan, Liang, Ma (2022, “An Explanation of In-context Learning as Implicit Bayesian Inference,” arXiv:2111.02080) argue that pretraining on data with latent “concepts” induces a posterior over concepts given the prompt; ICL is implicit posterior inference.

Implicit gradient descent. von Oswald et al. (2023, “Transformers Learn In-Context by Gradient Descent,” arXiv:2212.07677) show that trained transformers’ attention layers can be read as implementing one step of gradient descent on an inner regression problem. Several follow-ups have refined this.

Induction-head circuits. Olsson et al. identify induction heads as the mechanistic substrate of much in-context learning; ICL emerges empirically when induction heads form during training.

These are not exclusive. The Bayesian and gradient-descent views are abstractions of what may be implemented, in part, by induction-head-like circuits. For Noosphere, the relevance is that prompts for claim extraction are themselves a form of ICL — and the literature provides traction on why prompt-format details (number of examples, order of examples, formatting consistency) have large effects on output quality.

8.6 Chain-of-thought and faithfulness

Wei et al. (2022, “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” arXiv:2201.11903) showed that prompting a model to “think step by step” before giving a final answer improves performance on reasoning tasks.

The critical follow-up: Turpin et al. (2023, “Language Models Don’t Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting,” arXiv:2305.04388). They demonstrated that LLM-generated reasoning is *not* always the actual cause of the final answer. Introducing biasing features in the prompt changes the answer but the model’s stated CoT does not acknowledge the bias.

For Noosphere. If a coherence judgement asks the LLM judge for a reason and stores that reason as part of the trace, the firm should not treat that reason as a faithful explanation of the model’s verdict. Anthropic’s own “Tracing the Thoughts” work (Lindsey et al., 2025, discussed in Part VII) finds explicit cases of post-hoc rationalization at the activation level.

Sources to read in this group:

- Wei et al., *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*, NeurIPS 2022. arXiv:2201.11903.
- Turpin et al., *Language Models Don’t Always Say What They Think*, NeurIPS 2023. arXiv:2305.04388.
- Wang et al., *Self-Consistency Improves Chain-of-Thought Reasoning*, ICLR 2023. arXiv:2203.11171.
- Saurav Kadavath et al., *Language Models (Mostly) Know What They Know*, Anthropic 2022. arXiv:2207.05221. Foundational on LLM self-assessment of confidence.

8.7 The alignment stack: SFT, RLHF, DPO, Constitutional AI

Supervised fine-tuning (SFT) trains on demonstrations (x, y) with cross-entropy loss. This is what makes a base model into an instruction follower at all.

RLHF (Christiano et al., 2017; Stiennon et al., 2020; Ouyang et al., 2022) consists of three stages: SFT; train a reward model $r_\phi(x, y)$ on human preference data using a Bradley–Terry log-likelihood loss; optimize the policy π_θ with PPO against r_ϕ plus a KL penalty against a reference policy π_{ref} :

$$\mathcal{L}_{\text{RLHF}} = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} [r_\phi(x, y)] - \beta D_{\text{KL}}(\pi_\theta(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The KL penalty prevents the policy from drifting too far from the base model and “reward hacking” the imperfect r_ϕ .

Direct Preference Optimization (Rafailov et al., 2023, arXiv:2305.18290) is a recent simplification. The RLHF KL-constrained optimum has a closed form, $\pi^*(y | x) \propto \pi_{\text{ref}}(y | x) \exp(r(x, y)/\beta)$, which rearranged gives $r(x, y) = \beta \log[\pi^*(y | x)/\pi_{\text{ref}}(y | x)] + Z(x)$. Substituting this into the Bradley–Terry preference likelihood eliminates the explicit reward model and turns RLHF into a direct supervised loss on π_θ . DPO is now widely adopted.

Constitutional AI (Bai et al., 2022, arXiv:2212.08073) replaces human-feedback labels with model-generated critiques against a principle list (a “constitution”), then trains on the model-revised outputs. RLAIF extends this to RL on AI-generated preference data.

For Noosphere. The Claude model used in Noosphere has been through SFT, RLHF or DPO, and Constitutional or RLAIF training. Its priors include helpfulness, harmlessness, refusal patterns, hedging, and stylistic conventions. A neutral observer is unavailable; Noosphere is reading the world through a model that has been deliberately shaped by these layers.

8.8 Inference: sampling, temperature, speculative decoding

Sampling. Autoregressive generation is parameterized by a sampling strategy. *Temperature* τ scales logits before softmax. *Top- k* restricts to the k highest-probability tokens. *Top- p* (*nucleus*, Holtzman et al. 2019) restricts to the smallest set of tokens whose cumulative probability exceeds p .

Greedy/argmax ($\tau \rightarrow 0$) is deterministic but not truth-preserving — the model can be deterministically wrong if its prior is wrong. For extraction tasks, deterministic decoding is the safer choice; for generation tasks where diversity matters, sampling is better.

Constrained decoding. When the output must conform to a grammar (JSON schema, regular expression), the decoder can mask out tokens that would violate the grammar at each step. Outlines (Willard & Louf, 2023, arXiv:2307.09702) and the Grammars-Forms apparatus implement this. For Noosphere’s JSON-schema-bound claim extraction, constrained decoding is the correct primitive.

Speculative decoding (Leviathan et al., 2023, arXiv:2211.17192): a small “draft” model proposes several tokens; the large model verifies in parallel; accepted prefixes are committed. Reduces wall-clock latency by 2–3x without changing the output distribution.

9 The Residual Stream Perspective

Anthropic’s mathematical framework (Elhage et al. 2021) reframes the transformer as a sequence of *additive* updates to a single shared representation, the “residual stream.” At each layer ℓ and

position t :

$$x_t^{(\ell+1)} = x_t^{(\ell)} + \text{Attn}_\ell(x_{1:T}^{(\ell)})_t + \text{MLP}_\ell(x_t^{(\ell)} + \text{Attn}_\ell(x_{1:T}^{(\ell)})_t).$$

Because all updates are additive, the residual stream at any layer is a sum of contributions from each preceding component (token embedding, every attention head at every prior layer, every MLP). This linear decomposition is the analytical backbone of mechanistic interpretability: you can attribute the final logit for token y to specific attention heads or MLPs by tracking which components write into the relevant direction of the residual stream.

For Noosphere. If you wanted to ask, mechanistically, “which attention heads in Claude are doing the work of recognizing a contradiction in this sentence pair,” the residual-stream decomposition is the framework in which that question becomes well-posed. The tools (activation patching, path patching, attribution graphs — Part VII) operate on this picture.

10 Recommended reading for Part II

- Jianlin Su et al., *RoFormer: Enhanced Transformer with Rotary Position Embedding*, arXiv:2104.09864, 2021.
- Ofir Press, Noah A. Smith, Mike Lewis, *Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation* (ALiBi), ICLR 2022. arXiv:2108.12409.
- Sang Michael Xie et al., *An Explanation of In-context Learning as Implicit Bayesian Inference*, ICLR 2022. arXiv:2111.02080.
- Johannes von Oswald et al., *Transformers Learn In-Context by Gradient Descent*, ICML 2023. arXiv:2212.07677.
- Rafael Rafailov et al., *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*, NeurIPS 2023. arXiv:2305.18290.
- Long Ouyang et al., *Training Language Models to Follow Instructions with Human Feedback* (InstructGPT), 2022. arXiv:2203.02155.
- Yuntao Bai et al., *Constitutional AI: Harmlessness from AI Feedback*, 2022. arXiv:2212.08073.
- Ari Holtzman et al., *The Curious Case of Neural Text Degeneration* (nucleus sampling), ICLR 2020. arXiv:1904.09751.
- Yaniv Leviathan et al., *Fast Inference from Transformers via Speculative Decoding*, ICML 2023. arXiv:2211.17192.
- Brandon T. Willard & Rémi Louf, *Efficient Guided Generation for Large Language Models* (Outlines), arXiv:2307.09702, 2023.
- Nelson Elhage et al., *A Mathematical Framework for Transformer Circuits*, Anthropic 2021. <https://transformer-circuits.pub/2021/framework/index.html>.

Part III — Embedding Geometry: The Full Apparatus

11 Static vs. Contextual Embeddings

Word2vec and GloVe produce *static* embeddings: each word type has one vector regardless of context. Contextual embeddings (BERT, GPT, Sentence-BERT, modern API embeddings from OpenAI/Voyage/Cohere/BGE) produce context-dependent vectors. The difference matters because logical operations on text — negation, scope, hedging, attribution — are exactly the operations that static embeddings cannot represent and contextual embeddings must.

The cost of contextuality is that the same word now occupies many positions in space, related but not identical. “Bank” in “river bank” and “bank” in “money bank” produce different vectors; this is desirable for word-sense disambiguation but complicates any analysis that treats words as having a single “meaning location.”

11.1 Pooling: how a span becomes a vector

A multi-token span has no single canonical embedding. Common choices: (a) the embedding of a special [CLS] or [SEP] token; (b) mean of token embeddings; (c) max-pool; (d) last-token embedding; (e) attention-weighted pooling with a learned query. Sentence-BERT uses mean pooling on [mean,max,cls] concatenations, fine-tuned with a contrastive loss.

This is a non-trivial design decision. A coherence engine that compares “claim” embeddings is comparing pooled vectors, and the pool’s failure modes (mean pooling diluting key tokens, max pooling overweighting outliers) propagate into the score.

12 Anisotropy and Its Remedies, Quantified

Section 2.3 of Part I introduced anisotropy. Here is the quantitative treatment.

12.1 Measuring anisotropy

Let $\{\mathbf{x}_i\}_{i=1}^N$ be embeddings. The mean-cosine-similarity between random pairs, $\bar{c} = \frac{1}{N(N-1)} \sum_{i \neq j} \cos(\mathbf{x}_i, \mathbf{x}_j)$, is a zeroth-order measure: for isotropic data $\bar{c} \rightarrow 0$; values close to 1 indicate severe anisotropy. Better: the ratio of the second to the first singular value of the centered embedding matrix $\tilde{X} = X - \mathbf{1}\bar{\mathbf{x}}^\top$. For perfectly isotropic data the singular values are roughly equal; for cone-shaped data the first singular value dominates.

12.2 Remedies

All-but-the-top (Mu & Viswanath): center the embeddings, then remove the top- d' principal components. Empirically $d' \approx d/100$ works.

Whitening: $\mathbf{x}_i \mapsto W(\mathbf{x}_i - \bar{\mathbf{x}})$ with $W = U\Lambda^{-1/2}$, where $U\Lambda U^\top$ is the eigendecomposition of the empirical covariance. After whitening, components are decorrelated and unit-variance.

Sentence-BERT supervision: contrastive training explicitly pushes unrelated pairs apart, which removes much of the anisotropy without an explicit post-hoc step.

For Noosphere. If your embedding model produces highly anisotropic vectors and you use raw cosine, you are not measuring semantic similarity — you are measuring how much each pair conforms to the dominant direction. Whitening before any geometric computation is the cheapest correction.

13 Probes Done Right

A *probe* is a small (usually linear) classifier trained on top of frozen representations to predict a property. The probe’s accuracy is taken as evidence that the representation encodes the property. This is the methodological backbone of the linear-representation-hypothesis literature and the foundation of CCS, Geometry of Truth, and inference-time intervention.

13.1 The Hewitt–Liang controls

Hewitt & Liang (2019, “Designing and Interpreting Probes with Control Tasks,” arXiv:1909.03368) gave a now-canonical methodology: every probing experiment must include a *control task* where the labels are random but consistent. *Selectivity* is the gap between the probe’s accuracy on the real task and on the control task. High accuracy with low selectivity means the probe is memorizing, not reading off the representation.

For Noosphere. Any claim of the form “my probe finds a contradiction direction with accuracy A ” is weak evidence without an accompanying control: train the same probe on randomly assigned “contradicts/coheres” labels, and report the accuracy gap. If the gap is small, the probe is fitting noise, not reading geometry.

13.2 Causal abstraction and DAS

Geiger et al. (2021, “Causal Abstractions of Neural Networks,” arXiv:2106.02997; 2023, “Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations,” arXiv:2303.02536) develop the formal framework for asking “does this representation implement this causal structure.” Distributed Alignment Search (DAS) searches for a subspace in which a target causal variable is encoded. This is the rigorous upgrade of the linear-probe paradigm: instead of asking “can I predict the label,” you ask “is intervening on this subspace equivalent to intervening on the abstract variable in the desired causal model.”

13.3 Activation patching

Activation patching (also *causal mediation*): run the model on two inputs x_{base} and x_{src} ; replace a target activation at some layer/component when running on x_{base} with the corresponding activation from running on x_{src} ; observe whether the output changes toward x_{src} ’s prediction. The size of the shift quantifies the causal role of that activation in mediating the behavioral difference.

Vig et al. (2020, arXiv:2004.12265) introduced this in NLP for studying gender bias. Meng et al. (2022, ROME, arXiv:2202.05262) used it to localize factual associations. Conmy et al. (2023, ACDC, arXiv:2304.14997) automated circuit discovery using it. *Edge Attribution Patching* (Syed et al. 2023, arXiv:2310.10348; Hanna et al. 2024) is a linear approximation that scales the procedure to large models.

For Noosphere. The local contradiction engine asks a behavioral question: “does the geometry of this embedding pair signal contradiction?” The activation-patching toolkit answers the mechanistic version: “does intervening on this candidate direction *cause* the model to change its contradiction

judgement?” These are different questions. The behavioral question has a yes/no answer empirically; the mechanistic question has a yes/no answer that licenses much stronger claims about *what* the geometry is doing.

14 Sparse Autoencoders, Done Seriously

Volume I named sparse autoencoders. Here is the actual apparatus.

14.1 The dictionary-learning formulation

Given activations $\mathbf{x} \in \mathbb{R}^d$ (typically from the residual stream at a chosen layer), an SAE is an encoder–decoder pair:

$$\mathbf{f} = \sigma(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{x}} = W_{\text{dec}}\mathbf{f} + \mathbf{b}_{\text{pre}}.$$

The hidden dimension is *overcomplete*: $|\mathbf{f}| \gg d$, often $8\times$ to $32\times$. The loss is $\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{f}\|_1$. The ℓ_1 penalty encourages sparsity in $\mathbf{f}$, so each input activates only a small subset of dictionary features.

14.2 Architectural variants (2023–2025)

Vanilla ReLU SAEs have a bias problem: the ℓ_1 penalty pushes activations toward zero even when the underlying feature is present.

Top- k SAEs (Gao et al., OpenAI, 2024, arXiv:2406.04093): keep only the top- k activations and zero the rest. Cleanly decouples sparsity from ℓ_1 shrinkage.

Gated SAEs (Rajamanoharan et al., DeepMind, 2024, arXiv:2404.16014): separate the “gate” (which features fire) from the “magnitude” (how strongly), with two heads in the encoder.

JumpReLU SAEs (Rajamanoharan et al., DeepMind, 2024, arXiv:2407.14435): replace ReLU with a discontinuous JumpReLU activation; cleaner trade-off curves between reconstruction and sparsity.

Switch / Mixture of SAEs (Mudide et al., 2024, arXiv:2410.08201): multiple SAE experts, gated; substantially better at scale.

14.3 Crosscoders and transcoders

Crosscoders (Lindsey et al., Anthropic, 2024) jointly encode activations from multiple layers or multiple models, sharing a dictionary. Useful for finding features that persist across layers or generalize across model families.

Transcoders (Dunefsky et al., 2024) replace MLP blocks with sparse autoencoder-like maps; this gives a cleaner unit of analysis than “SAE on residual stream” because the transcoder *is* the computation.

14.4 Empirical findings

Scaling Monosemanticity (Templeton et al., Anthropic, 2024) applied SAEs to Claude 3 Sonnet and recovered millions of human-interpretable features, including features for specific people,

programming bugs, and emotional tones. Features can be *steered*: amplifying the “Golden Gate Bridge” feature made the model bring it up in every response.

Feature splitting: as SAE size increases, what was one feature at small scale tends to split into multiple more-specific features at larger scale. This is a methodological warning: the granularity of “features discovered” depends on the SAE configuration, not just on the model.

Limitations. SAEs reconstruct only part of the variance (typically 80–95%); the unreconstructed part is presumed to encode features the particular SAE missed. SAE features are also model- and layer-specific.

15 Attribution Graphs (Anthropic, March 2025)

The most significant recent development in mechanistic interpretability is the attribution-graph methodology described in Anthropic’s “On the Biology of a Large Language Model” (Lindsey et al., March 2025, <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>).

15.1 What an attribution graph is

An attribution graph traces, for a specific prompt and output, the chain of SAE features and attention edges that contributed to the output token. Nodes are features at specific (layer, token) positions; edges are estimated causal contributions. The result is a directed graph that can be inspected, pruned, and used to hypothesize circuit-level explanations.

15.2 Findings relevant to Noosphere

The 2025 paper documents several findings of direct relevance:

- **Two-hop reasoning “in the head.”** For “the capital of the state containing Dallas is ___”, the model represents “Texas” as an intermediate feature even though that token never appears in the prompt.
- **Planning in poetry generation.** Before generating a rhyming line, the model encodes the target rhyme word in advance.
- **Multilingual circuits.** Concept-level features are shared across languages; the model translates near the surface.
- **Post-hoc rationalization.** Cases are documented where the model’s stated reasoning differs from the actual computation revealed by the attribution graph.
- **Limitations.** The methodology yields a satisfying account for roughly a quarter of attempted prompts; for the rest, the graph is too diffuse to interpret confidently.

For Noosphere. Attribution graphs are the most ambitious available attempt to make LLM computation auditable at a level beyond input–output behavior. If Noosphere is serious about “trace as source of truth,” the research frontier is racing toward traces that go inside the model.

16 Recommended reading for Part III

- John Hewitt & Percy Liang, *Designing and Interpreting Probes with Control Tasks*, EMNLP 2019. arXiv:1909.03368.

- Atticus Geiger, Hanson Lu, Thomas Icard, Christopher Potts, *Causal Abstractions of Neural Networks*, NeurIPS 2021. arXiv:2106.02997.
- Atticus Geiger et al., *Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations* (DAS), 2023. arXiv:2303.02536.
- Jesse Vig et al., *Causal Mediation Analysis for Interpreting Neural NLP*, NeurIPS 2020. arXiv:2004.12265.
- Kevin Meng et al., *Locating and Editing Factual Associations in GPT (ROME)*, NeurIPS 2022. arXiv:2202.05262.
- Arthur Conmy et al., *Towards Automated Circuit Discovery for Mechanistic Interpretability* (ACDC), NeurIPS 2023. arXiv:2304.14997.
- Aaquib Syed, Can Rager, Arthur Conmy, *Attribution Patching Outperforms Automated Circuit Discovery*, 2023. arXiv:2310.10348.
- Trenton Bricken et al., *Towards Monosemanticity: Decomposing Language Models with Dictionary Learning*, Anthropic 2023. <https://transformer-circuits.pub/2023/monosemantic-features>.
- Leo Gao, Tom Dupré la Tour, Henk Tillman et al., *Scaling and Evaluating Sparse Autoencoders*, OpenAI 2024. arXiv:2406.04093.
- Senthooran Rajamanoharan et al., *Improving Dictionary Learning with Gated Sparse Autoencoders*, DeepMind 2024. arXiv:2404.16014.
- Senthooran Rajamanoharan et al., *JumpReLU Sparse Autoencoders*, DeepMind 2024. arXiv:2407.14435.
- Jacob Dunefsky et al., *Transcoders Find Interpretable LLM Feature Circuits*, NeurIPS 2024. arXiv:2406.11944.
- Jack Lindsey et al., *On the Biology of a Large Language Model*, Anthropic, March 2025. <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.

Part IV — NLP Extraction: Production-Grade Pipelines

17 Coreference at the State of the Art

Lee et al. (2017) introduced end-to-end neural coreference. The post-2020 state of the art has moved to higher-recall systems with explicit mention detection and clustering.

LingMess and Word-level coreference (Otmazgin, Cattan, Goldberg 2022, arXiv:2205.12644; Dobrovolskii 2021, arXiv:2109.04127): word-level rather than mention-level resolution improves both speed and accuracy.

Maverick (Martinelli et al., 2024, arXiv:2407.21489): the current public state of the art on OntoNotes; lightweight architecture, strong generalization.

LLM-based coreference: GPT-4-class models can do zero-shot coreference competitively (Le & Ritter 2023, arXiv:2310.14256). Hybrid approaches that ask an LLM to verify or merge clusters from a dedicated system often outperform either alone.

For Noosphere. The Dialectic component analyzing live conversation faces especially hard coreference: speakers, pronouns referring to abstract arguments (“that point”), and forward references (“what I’m about to say”). The right architecture stores low-confidence coreference links as candidates and surfaces them in the trace, rather than committing silently.

18 Entity Linking

Entity linking maps mentions to canonical entity identifiers (typically Wikidata QIDs or Wikipedia titles).

BLINK (Wu, Petroni, Josifoski, Riedel, Zettlemoyer, 2020, arXiv:1911.03814): bi-encoder retrieval + cross-encoder reranking, the canonical modern architecture.

GENRE (De Cao, Izacard, Riedel, Petroni, 2021, arXiv:2010.00904): generative entity retrieval; the model generates the entity title autoregressively, conditioned on the mention.

ReFinED (Ayoola et al., 2022): single-pass entity linking combining mention detection, type assignment, and disambiguation.

For Noosphere, the operational risk is the opposite of textbook entity linking: most domain-relevant entities (people in the firm’s network, proprietary concepts, internal projects) are *not* in Wikidata. The firm needs a custom knowledge-base linker over its own entity registry, with a fallback to general-purpose linkers for public entities.

19 Schema-Guided LLM Extraction

The Volume I description of **ClaimExtractor** as “JSON-bound” extraction is correct but understated. Three sharper techniques.

19.1 Constrained decoding

As discussed in Part II, constrained decoding masks the next-token distribution to enforce a grammar. For JSON: at each step, the decoder considers only tokens that lead to a string still extensible to a valid JSON instance of the target schema. Outlines, llguidance, and the `json_schema` mode of major API providers all implement this. With constrained decoding, schema violations become impossible; without it, they become rare but devastating bugs.

19.2 Few-shot vs. zero-shot, and the demonstration-selection problem

For sufficiently capable models, zero-shot extraction with a detailed schema and instructions outperforms few-shot on many tasks (Wadhwa et al. 2023, arXiv:2305.05003). Where few-shot helps, the choice of demonstrations matters disproportionately: relevant, diverse, format-consistent demonstrations help; unrelated or inconsistent ones can hurt.

19.3 Structured Prompt Engineering for extraction

The current production-grade pattern (also reflected in Anthropic’s prompt guidelines):

1. State the role and the contract (“You are an extractor. Output JSON matching this schema. Refuse with a specific error code if the text is unsuitable.”).
2. Define every field unambiguously, including edge cases.
3. Provide 2–5 carefully chosen worked examples with explicit edge cases (refused, hedged, multi-claim).
4. Provide the target text with clear delimiters.
5. Demand the output as the last thing in the response (no preamble).
6. Validate the output against a JSON schema; on validation failure, either re-prompt with the validator’s error message or escalate to a larger model.

20 Argument Mining: State of the Art

Volume I cited Stab & Gurevych. Recent work that is required reading if Noosphere wants to extract argument structure from text:

Lawrence & Reed (2020), “Argument Mining: A Survey,” Computational Linguistics 45(4). The standard recent survey.

Persing & Ng work on essay-level argumentation (multiple papers, ACL 2015 onward).

Bao et al. (2022), “Have my arguments been replied to?,” arXiv:2202.02613, ACL 2022 — a useful test of whether systems can find the structure of reply-and-counterargument.

IBM Project Debater corpus (Slonim et al., Nature 2021): the most serious end-to-end argument-mining application; the underlying datasets (claim-stance, argument-quality, evidence retrieval) are publicly available and are the closest extant analogues to Noosphere’s principle-distillation task.

Walton’s argumentation schemes: Douglas Walton (*Argumentation Schemes*, Cambridge 2008) catalogued 60+ recurring argument patterns (argument from expert opinion, argument from analogy, slippery slope, etc.) each with characteristic critical questions. Annotating texts against these

schemes is a richer task than “claim/premise/support” and is closer to what Noosphere’s principle extractor should ultimately produce.

21 Stance Detection and Discourse Parsing

Stance detection: given a text and a target proposition, classify the text’s stance toward the target as **pro** / **con** / **neutral** / **unrelated**. The SemEval-2016 Task 6 dataset is the standard benchmark. Hardalov et al. (2022, “Few-Shot Cross-Lingual Stance Detection,” arXiv:2106.06632) is a recent strong system.

Discourse parsing: Two competing frameworks. *Rhetorical Structure Theory* (RST; Mann & Thompson 1988) builds trees of nucleus–satellite relations (Elaboration, Contrast, Cause, etc.). *Penn Discourse Treebank* (PDTB; Prasad et al. 2008) annotates discourse connectives and their arguments, without committing to a tree structure.

For Noosphere, discourse-level structure tells you which sentences belong to the same argument unit, which are evidence vs. thesis, and where in a transcript the reasoning is happening. A claim extractor that ignores discourse markers will conflate the speaker’s argument with the position they are entertaining only to refute it.

22 Evaluation Methodology for Extraction

22.1 Metrics, properly

Precision, recall, F1: standard. For span-level tasks (NER, claim extraction), exact-match F1 and partial-overlap F1 give different pictures.

B³, MUC, CEAF, BLANC: coreference metrics. They reward different properties (entity-aligned vs. mention-aligned). Reporting only one is incomplete; the CoNLL-2012 standard is the average of MUC, B³, and CEAF_{e4}.

Slot filling F1: for structured extraction with multiple fields, average per-field F1 is informative; macro-averaging weights all entity types equally.

Hallucination rate: for LLM extractors, the fraction of extracted items not supported by the source. The QAGS and FEVER methodologies are the standard. Min et al. (2023, FActScore, arXiv:2305.14251) is the recent strong evaluator.

22.2 Annotator agreement

Cohen’s κ and Krippendorff’s α measure inter-annotator agreement beyond chance. For tasks where the “ground truth” is itself ambiguous (argument mining, stance), high IAA is hard and any reported F1 should be calibrated against it. A model F1 of 0.78 on a task with human IAA of 0.65 is at the human ceiling.

22.3 Recommended reading for Part IV

- John Lawrence & Chris Reed, *Argument Mining: A Survey*, Computational Linguistics 45(4), 2020. <https://aclanthology.org/J19-4006/>.
- Douglas Walton, Chris Reed, Fabrizio Macagno, *Argumentation Schemes*, Cambridge University Press 2008.

- Noam Slonim et al., *An autonomous debating system*, Nature 591, 2021.
- Sebastian Hofstätter, Sophia Althammer et al., *ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction*, NAACL 2022. arXiv:2112.01488.
- Sewon Min et al., *FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation*, EMNLP 2023. arXiv:2305.14251.
- Patrick Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, NeurIPS 2020. arXiv:2005.11401.
- Akari Asai et al., *Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection*, ICLR 2024. arXiv:2310.11511.

Part V — Graphs, Ontologies, and Formal Logic

23 RDF, OWL, SPARQL: a Working Vocabulary

If Noosphere’s ontology ever moves beyond a property graph, it will move into the Semantic Web stack. The minimum vocabulary:

RDF (Resource Description Framework): triples (s, p, o) where s is a subject, p a predicate, o an object. Subjects and predicates are IRIs (or blank nodes for subjects); objects are IRIs or literals. A graph is a set of triples. *RDF Schema (RDFS)* adds basic vocabulary: `rdfs:subClassOf`, `rdfs:subPropertyOf`, `rdfs:domain`, `rdfs:range`.

OWL (Web Ontology Language): a more expressive ontology language built on RDF. OWL 2 has three computationally tractable *profiles* — OWL 2 EL, OWL 2 QL, OWL 2 RL — each with polynomial-time reasoning for different fragments, plus the full OWL 2 DL which is decidable but 2NEXPTIME-complete.

SPARQL: the query language. Pattern-matches triples; supports joins, optional patterns, unions, federated queries, and aggregations.

SHACL (Shapes Constraint Language): separates validation constraints from the ontology proper. Defines “shapes” that nodes must satisfy. For Noosphere’s claim and principle schemas, SHACL is the right formalism for the assertions you currently encode in Pydantic.

24 Description Logic Formally

OWL is a syntactic surface for the description logic *SROIQ*. The core ideas:

24.1 ALC: the base description logic

ALC (Attribute Language with Complement) has concepts (sets of individuals), roles (binary relations), and the constructors:

$$C, D ::= A \mid \top \mid \perp \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \exists r.C \mid \forall r.C.$$

Here A is a primitive concept, r is a role, $\exists r.C$ is the class of individuals with at least one r -successor in C , $\forall r.C$ requires all r -successors in C .

A **TBox** is a set of *terminological axioms* (subsumptions $C \sqsubseteq D$, equivalences $C \equiv D$). An **ABox** is a set of *assertions* ($a : C$ saying individual a is in concept C , or $(a, b) : r$ saying a is r -related to b).

Two inference services dominate. **Subsumption:** given a TBox $\mathcal{T}$, is $\mathcal{T} \models C \sqsubseteq D$? **Instance checking:** given $\mathcal{T}, \mathcal{A}$, does the individual a belong to C ?

24.2 SROIQ and OWL 2 DL

SROIQ extends ALC with role hierarchies, inverse roles, transitive roles, nominals (single-individual classes $\{a\}$), and qualified cardinality restrictions ($\leq nr.C$, $\geq nr.C$). OWL 2 DL is the syntactic surface of SROIQ with restrictions ensuring decidability. Reasoning is 2NEXPTIME-complete in

the worst case but practical implementations (Pellet, HermiT, FaCT++, Konclude) handle realistic ontologies.

24.3 OWL 2 profiles

OWL 2 EL: polynomial-time reasoning. Supports existential restrictions and intersections; omits negation, disjunction, universal restrictions. The basis of biomedical ontologies (SNOMED CT, the Gene Ontology). Reasoner: ELK.

OWL 2 QL: designed for query answering over relational databases. Reasoning reducible to first-order rewriting.

OWL 2 RL: designed to be implementable as forward-chaining rules in standard rule engines. The right choice if Noosphere wants to implement ontology-based inference inside a property-graph store.

24.4 Open vs. closed world

DL reasoning is *open world*: failure to find evidence for $C(a)$ does not mean $\neg C(a)$. Most operational reasoning (databases, Prolog) is *closed world*. For a firm's intellectual capital, the open-world assumption is the correct default: the absence of an asserted contradiction is not a guarantee that none exists.

25 Belief Revision: AGM Postulates

Alchourrón, Gärdenfors, Makinson (1985, JSL 50(2), 510–530) gave the classical axiomatization of rational belief change. Let K be a belief set (deductively closed); $\dot{+}$ contraction (removing belief), $*$ revision (adding new belief consistently), $+$ expansion (adding without consistency check).

25.1 The AGM postulates for revision

- (K*1) $K * \phi$ is a belief set (closure).
- (K*2) $\phi \in K * \phi$ (success).
- (K*3) $K * \phi \subseteq K + \phi$ (inclusion).
- (K*4) If $\neg\phi \notin K$ then $K + \phi \subseteq K * \phi$ (preservation).
- (K*5) $K * \phi$ is inconsistent only if ϕ is inconsistent (consistency).
- (K*6) If $\vdash \phi \leftrightarrow \psi$ then $K * \phi = K * \psi$ (extensionality).
- (K*7) $K * (\phi \wedge \psi) \subseteq (K * \phi) + \psi$ (superexpansion).
- (K*8) If $\neg\psi \notin K * \phi$ then $(K * \phi) + \psi \subseteq K * (\phi \wedge \psi)$ (subexpansion).

25.2 Why this matters for Noosphere

The Theseus belief layer must support retraction: a source recants, a forecast resolves contrary to a principle, an internal review identifies a flaw. AGM-compliant revision tells you what changes: minimally, in a way consistent with the new information. The Levi identity $K * \phi = (K \dot{-} \neg\phi) + \phi$ links revision to contraction; implementing one gives you the other.

The classical AGM framework is for monotonic logical closure under classical entailment. Noosphere’s graph is closer to a non-monotonic system (principles override default extrapolations from claims). Reiter’s default logic, Pollock’s defeasible reasoning, and the literature on truth maintenance systems (Doyle 1979; de Kleer 1986, ATMS) are the relevant extensions.

26 Dung’s Argumentation Frameworks

Phan Minh Dung (1995, “On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games,” *Artificial Intelligence* 77, 321–357) formalized abstract argumentation. The framework is shockingly simple and remarkably useful.

26.1 Definition

An *argumentation framework* (AF) is a pair $(\mathcal{A}, \mathcal{R})$ where $\mathcal{A}$ is a set of arguments and $\mathcal{R} \subseteq \mathcal{A} \times \mathcal{A}$ is an *attack* relation. That is all. No internal structure of arguments is assumed.

A set $S \subseteq \mathcal{A}$ is *conflict-free* if no $a \in S$ attacks any $b \in S$. S *defends* a if every attacker of a is attacked by some member of S . S is *admissible* if it is conflict-free and defends all its members.

26.2 Semantics (extensions)

Several notions of “which sets of arguments are jointly acceptable”:

- *Complete*: admissible and contains every argument it defends.
- *Grounded*: the smallest complete extension. Always unique.
- *Preferred*: a maximal admissible extension.
- *Stable*: a conflict-free set that attacks every argument not in it.

For Noosphere. The cascade graph’s claims with `contradicts` edges naturally form an AF. Computing the grounded extension gives the “unambiguous” beliefs under the current evidence; computing preferred extensions gives the maximal coherent positions. The skeptical (intersection of all preferred extensions) vs. credulous (union) acceptability of an argument are formal counterparts of the firm’s “definitive vs. provisional” verdicts.

26.3 Beyond Dung: ASPIC⁺ and bipolar AFs

ASPIC⁺ (Modgil & Prakken 2014) adds internal argument structure: arguments have premises and inference rules; attacks decompose into undercut (attacks on inference rule), rebuttal (attacks on conclusion), and undermining (attacks on premise). Bipolar AFs add a *support* relation alongside attack, with several proposals for combined semantics (Cayrol & Lagasque-Schiex 2005, Amgoud et al. 2008).

For Noosphere. The cascade graph has both supports and contradicts edges, so bipolar AFs are the natural formalization. ASPIC⁺ is the right formalism if you want to record internal argument structure (premises, inference rules) alongside the attack relation.

27 Probabilistic Graphical Models, Properly

27.1 Bayesian networks vs. Markov random fields

A *Bayesian network* is a DAG with conditional probability tables. A *Markov random field* (Markov network) is an undirected graph with potential functions over cliques. BNs are natural for causal/temporal modeling; MRFs are natural when symmetric relations (similarity, attraction, inhibition) are primary.

For Noosphere’s cascade graph, BNs are the better fit because the dominant edges (supports, derives-from, refines) are directional. The MRF formalism becomes relevant for symmetric “coherence” relations.

27.2 Inference: exact and approximate

Exact inference in a BN is sum-product over the joint, which is #P-hard in general. Practical exact methods: *variable elimination* (quadratic in the largest factor), *junction trees* (tree decomposition), *message passing* on polytrees (linear).

Approximate inference: *Markov Chain Monte Carlo* (Gibbs, Metropolis–Hastings), *variational inference* (mean-field, structured), *loopy belief propagation*.

For Noosphere. The cascade graph projected to a Bayesian skeleton is small enough that exact inference is feasible; for larger graphs, loopy belief propagation is the easiest practical choice but has no convergence guarantees on graphs with cycles. The choice should be documented and versioned in the trace.

27.3 Plate notation and parameter sharing

A *plate* in a graphical-model diagram denotes a repeated subgraph with shared parameters. Plate notation is the standard way to draw parameter-shared models (LDA, hierarchical models). For Noosphere, the plate-notation question is which parameters are shared across claims of the same type (*e.g.*, does every “predicts” edge use the same CPT, or do they differ by domain?).

28 Markov Logic, ProbLog, Probabilistic Soft Logic

Three frameworks for combining logic and probability:

Markov Logic Networks (Richardson & Domingos, 2006, “Markov Logic Networks,” *Machine Learning* 62, 107–136): weighted first-order formulas; each grounding contributes its weight to an MRF over ground atoms. A weight w means satisfying the formula is e^w times more likely. MLNs unify deductive and probabilistic reasoning; learning is hard (weight learning, structure learning).

ProbLog (De Raedt, Kimmig, Toivonen, 2007): probabilistic extension of Prolog. Each fact has a probability; the probability of a query is the sum over all proofs, weighted by their conjunction probabilities. The distribution-semantics framework.

Probabilistic Soft Logic (Bach et al., 2017, “Hinge-Loss Markov Random Fields and Probabilistic Soft Logic,” *JMLR* 18): continuous-valued relaxations of first-order rules; inference is convex optimization rather than #P-hard counting. Scales better than MLNs on large graphs.

For Noosphere. If the coherence engine is ever to formalize “weight of evidence” for graph-derived conclusions, one of these three is the right formalism. PSL is the most tractable; MLNs the most

expressive; ProbLog sits between. None replaces the LLM-judge layer, but any of them gives the coherence aggregator a sounder foundation than 4-of-6 **supermajority**.

29 Causal Graphs and Do-Calculus

Judea Pearl’s framework (*Causality*, Cambridge 2009) distinguishes *observational* from *interventional* from *counterfactual* queries. A *structural causal model* (SCM) is a set of equations $V_i := f_i(\text{Pa}(V_i), U_i)$ with exogenous noise U_i , plus a distribution over noise variables. The associated DAG is the causal graph.

The *do-operator* $\text{do}(X = x)$ denotes intervention — forcing X to value x regardless of its parents. $P(Y \mid \text{do}(X = x))$ is generally not equal to $P(Y \mid X = x)$. The *do-calculus* is a sound and (for atomic interventions) complete system of three rules for manipulating expressions involving do , in terms of separation properties of the causal graph.

29.1 Identifiability

Given observational data and a causal graph, when is a causal effect identifiable? Tian & Pearl’s ID algorithm (2002, AAAI) is a complete algorithm for atomic-intervention identifiability. Shpitser & Pearl (2006) extended this to conditional interventions.

For Noosphere. The principle “X supports Y” has both a correlational and an interventional reading. “Documents that endorse X tend to endorse Y” is correlational. “Adopting X causes one to come to believe Y” is interventional. The cascade graph as currently designed records the structural claim but does not distinguish these readings. If the firm wants its forecasts to be causal, the relevant theory is Pearl’s, and the data structures need to support do-calculus queries (or accept that they don’t, and stop using causal language).

29.2 Causal discovery

PC algorithm (Spirtes & Glymour 1991): the original constraint-based causal discovery algorithm. *GES* (Chickering 2002): score-based. *NOTEARS* (Zheng et al. 2018, arXiv:1803.01422): continuous optimization for DAG discovery. *LiNGAM* (Shimizu et al. 2006): exploits non-Gaussianity for identifiability.

These tools are slowly making their way into LLM/embedding research. For Noosphere, they are the formal counterpart to “how do I know X really supports Y, rather than both being effects of a common cause Z.”

30 Recommended reading for Part V

- Franz Baader, Diego Calvanese, Deborah McGuinness, Daniele Nardi, Peter Patel-Schneider (eds.), *The Description Logic Handbook*, 2nd ed., Cambridge 2007.
- Ian Horrocks et al., *OWL 2 Web Ontology Language Primer*, W3C 2012. <https://www.w3.org/TR/owl2-primer/>. Free.
- Carlos Alchourrón, Peter Gärdenfors, David Makinson, *On the logic of theory change: Partial meet contraction and revision functions*, Journal of Symbolic Logic 50(2), 1985.

- Sven Ove Hansson, *A Textbook of Belief Dynamics*, Kluwer 1999. The textbook treatment of AGM.
- Phan Minh Dung, *On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games*, Artificial Intelligence 77, 1995.
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- Daphne Koller & Nir Friedman, *Probabilistic Graphical Models: Principles and Techniques*, MIT Press 2009.
- Matthew Richardson & Pedro Domingos, *Markov Logic Networks*, Machine Learning 62, 2006.
- Stephen H. Bach et al., *Hinge-Loss Markov Random Fields and Probabilistic Soft Logic*, JMLR 18, 2017. arXiv:1505.04406.
- Judea Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge 2009.
- Judea Pearl, Madelyn Glymour, Nicholas P. Jewell, *Causal Inference in Statistics: A Primer*, Wiley 2016. The accessible introduction.
- Peter Spirtes, Clark Glymour, Richard Scheines, *Causation, Prediction, and Search*, 2nd ed., MIT Press 2001. Free at the authors' website. The reference for causal discovery.

Part VI — Real Epistemology: Coherence, Truth, and the Probabilistic Impossibility Results

31 Theories of Truth, Compared

Volume I gestured at the distinction between coherence-as-justification and coherence-as-truth. The literature is more developed than that.

31.1 Tarski’s semantic conception

Alfred Tarski (1933/1944) gave the formal semantic theory: “ ϕ is true” iff ϕ , where the right-hand ϕ is interpreted in a metalanguage strictly stronger than the object language. The famous schema:

“snow is white” is true if and only if snow is white.

This is not a substantive theory of truth so much as an adequacy condition: any theory of truth must satisfy the T-schema for the object language. Tarski showed that no consistent language can contain its own truth predicate (the indefinability theorem).

For Noosphere. A coherence engine that calls itself a “truth” engine is making the strong claim that coherence *is* truth, or at least implies it. Volume I correctly flagged this. The Tarski framework tells us why “truth” is a tricky predicate: it cannot be reduced to properties of sentences within a single language without paradox. Noosphere should reserve “true” for what is verified by the world (forecasts that resolved correctly, claims that withstood adversarial review) and use “coherent” for the geometric/graph property.

31.2 Correspondence, coherence, pragmatic, deflationary

Correspondence: a proposition is true iff it corresponds to a fact in the world. The classical default; descendants in Russell, Wittgenstein (*Tractatus*), David Armstrong (truthmaker theory).

Coherence: a proposition is true iff it coheres with a maximally inclusive coherent set. Bradley, Blanshard, BonJour (early). Difficulty: which set?

Pragmatic: truth is what works, or what inquiry would converge on in the long run. Peirce, James, Dewey, Rorty.

Deflationary: “true” is a logical device for endorsement, not a substantive predicate. Frank Ramsey, Paul Horwich’s “minimalism.” Truth is just disquotation: to say ϕ is true is no more than to assert ϕ .

For a firm whose product is methodology, the question to settle in writing is which theory of truth governs the firm’s claims. Most operational research practice presupposes correspondence (we are trying to find out what the world is like) with deflationist hygiene about how we talk (“true” is a thin label, not a separate property). Noosphere’s geometric hypothesis is then a *conjecture about a correlate of correspondence*, not a redefinition of truth.

32 Probabilistic Coherence Measures

The literature on quantitative coherence is technical and the results are counter-intuitive. The following items are required to make the six-layer engine defensible.

32.1 Shogenji’s measure

Shogenji (1999, “Is Coherence Truth-Conducive?”, Analysis 59) proposed the measure

$$C_S(\phi_1, \dots, \phi_n) = \frac{P(\phi_1 \wedge \dots \wedge \phi_n)}{P(\phi_1) \dots P(\phi_n)}.$$

$C_S > 1$ means positive dependence; $C_S = 1$ independence; $C_S < 1$ negative dependence. Simple, but does not behave well as n varies and ignores the strength of the dependence.

32.2 Olsson’s, Fitelson’s, and Glass’s measures

Olsson (2002): the relative overlap measure, $C_O = P(\phi_1 \wedge \dots \wedge \phi_n) / P(\phi_1 \vee \dots \vee \phi_n)$. Bounded in $[0, 1]$, 1 iff all ϕ_i are equivalent given consistency.

Fitelson (2003): an averaged confirmation measure across all ordered pairs of subsets, using a chosen confirmation measure.

Glass (2002): essentially the same as Olsson independently.

These measures are not equivalent. A worked example can have one ordering under Shogenji and the opposite ordering under Olsson.

32.3 The Bovens–Hartmann impossibility result

Bovens & Hartmann (*Bayesian Epistemology*, Oxford 2003) proved a result that should rule out naive coherence claims forever.

Result (Bovens–Hartmann)

There is no probabilistic coherence measure C such that, for all information sets S and S' and all individually equally credible witnesses, $C(S) > C(S')$ implies that the posterior probability of S ’s conjunction (given the witness reports) exceeds the posterior of S' ’s.

In ordinary language: no single coherence measure orders all information sets in a way that tracks posterior probability of joint truth. The “truth-conduciveness” of coherence is at best partial and context-dependent.

Olsson (*Against Coherence: Truth, Probability, and Justification*, Oxford 2005) gives an independent and converging negative result for a class of coherence orderings.

For Noosphere. A coherence score, however defined, cannot in general be read as a probability of truth. The six-layer engine produces a number; whatever the number is, it is not (without further justification) equal to $P(\text{true} \mid \text{evidence})$. The number can be useful for *ranking* candidate beliefs against each other under specific assumed witness structures, but those structures must be specified, and the ordering will not be invariant across measure choices.

32.4 What survives

Three rescues partially survive the impossibility results:

1. *Witness coherence*: under specific assumptions about the reliability and independence of sources, more coherent reports can be more probable. Bovens & Hartmann formalize this.
2. *Coherentism as relative*: coherence as a contributing justificatory factor, not as the sole one. Susan Haack's *foundherentism* (*Evidence and Inquiry*, Blackwell 1993) is the most worked-out hybrid.
3. *Process reliabilism plus coherence*: Goldman's externalist account makes justification a function of the reliability of the belief-forming process, supplemented by coherence for second-order defeat.

32.5 Bayesian confirmation theory in brief

Even without invoking coherence, one needs a theory of how evidence confirms hypotheses. The standard *ratio measure* $P(\phi \mid E)/P(\phi)$ and the *difference measure* $P(\phi \mid E) - P(\phi)$ disagree on rankings; the *log-likelihood ratio* $\log[P(E \mid \phi)/P(E \mid \neg\phi)]$ is the Bayesian-update log-odds shift and is widely preferred. The Hempel paradox of the ravens and Goodman's new riddle of induction (the famous "grue" problem) are the standard tests that any confirmation measure must address.

33 Inference to the Best Explanation

Charles Sanders Peirce's *abduction* and Gilbert Harman's (1965) *inference to the best explanation* (IBE) are the canonical names for the inferential move from evidence to a causal/structural hypothesis that explains it. Peter Lipton's *Inference to the Best Explanation* (2nd ed., Routledge 2004) is the standard book-length treatment.

Bas van Fraassen's critique (*Laws and Symmetry*, Oxford 1989, chap. 6): IBE adds nothing to Bayesian conditionalization, and may add illegitimate "best of a bad lot" bias if the search space is impoverished.

For Noosphere. The cascade graph's habit of identifying a "best-supported principle" for a given decision is implicitly IBE. The firm should be aware of (a) the dependence on the set of hypotheses considered (a bad lot can have a best member that is still wrong); (b) the risk of underweighting prior probability when "explanatory" goodness is emphasized; (c) the open question of how IBE relates to Bayesian updating, on which the literature is divided.

34 Reliabilism, Foundherentism, and the Internalism/Externalism Debate

Process reliabilism (Alvin Goldman, *Epistemology and Cognition*, Harvard 1986; *Reliabilism and Contemporary Epistemology*, Oxford 2012): a belief is justified iff it is produced by a reliable process. Justification is a function of the process, not (only) of the believer's reasons. This is *externalist*: facts about justification need not be accessible to the believer.

Foundherentism (Susan Haack): justification has both a foundational component (basic beliefs supported by experience) and a coherentist component (mutual support among non-basic beliefs). Beliefs are like crossword puzzle entries: supported partly by clues (foundations) and partly by interlocking with other entries (coherence).

Virtue epistemology (Ernest Sosa, *A Virtue Epistemology*, Oxford 2007; Linda Zagzebski, *Virtues of the Mind*, Cambridge 1996): knowledge is true belief produced by intellectual virtues (carefulness,

open-mindedness, sensitivity to evidence). The unit of evaluation shifts from beliefs to belief-forming agents.

For Noosphere. The firm’s intellectual capital model is closest in spirit to a process-reliabilist account: the firm’s principles are licensed not by the foundation they rest on but by the reliability of the process that produced them. This commits the firm to maintaining process metrics (forecast calibration, ablation-tested method versions, replication on new domains) at least as carefully as it maintains its principle library.

35 Recommended reading for Part VI

- Luc Bovens & Stephan Hartmann, *Bayesian Epistemology*, Oxford 2003. The reference for probabilistic coherentism and the impossibility results.
- Erik J. Olsson, *Against Coherence: Truth, Probability, and Justification*, Oxford 2005.
- Susan Haack, *Evidence and Inquiry: Towards Reconstruction in Epistemology*, Blackwell 1993.
- Alvin Goldman, *Reliabilism and Contemporary Epistemology*, Oxford 2012.
- Peter Lipton, *Inference to the Best Explanation*, 2nd ed., Routledge 2004.
- Bas C. van Fraassen, *Laws and Symmetry*, Oxford 1989. Chapter 6 is the IBE critique.
- Paul Horwich, *Truth*, 2nd ed., Oxford 1998. The deflationist reference.
- Donald Davidson, *Inquiries into Truth and Interpretation*, 2nd ed., Oxford 2001. The standard collection.
- James A. Hampton & Branden Fitelson, *Bayesian Confirmation Theory*, in *Continuum Companion to the Philosophy of Science*, 2011. A good entry to the confirmation-measure debate.
- Wesley C. Salmon, *Causality and Explanation*, Oxford 1998. For the explanation half of the IBE picture.

Part VII — The Frontier (2023–2026)

This part surveys research that is too recent to have settled, but too important to ignore. Each subsection is a portal; the cited papers are the correct entry points.

36 Sparse Autoencoders and What They Have Found

The SAE pipeline introduced in Part III is the central tool of frontier interpretability. Beyond *Towards Monosemanticity* (Bricken et al. 2023) and *Scaling Monosemanticity* (Templeton et al. 2024), the following developments matter.

Methodological progress. Top- k SAEs (Gao et al. 2024, arXiv:2406.04093), Gated SAEs (Rajamanoharan et al. 2024, arXiv:2404.16014), JumpReLU SAEs (Rajamanoharan et al. 2024, arXiv:2407.14435), and Switch SAEs (Mudide et al. 2024, arXiv:2410.08201) successively improved the reconstruction-sparsity Pareto frontier. The Anthropic and DeepMind teams ran a public bake-off (Karvonen et al. 2024) using interpretability-targeted metrics rather than just reconstruction loss.

Crosscoders (Lindsey et al., Anthropic 2024, <https://transformer-circuits.pub/2024/crosscoders/index.html>) jointly encode activations from multiple model layers (or multiple models), allowing the discovery of features that persist or diverge across layers/models. Particularly relevant for studying whether “the same concept” has the same representation at different depths.

Transcoders (Dunefsky et al. 2024, arXiv:2406.11944) replace MLP blocks with sparse-coded approximations; this gives a cleaner basis for circuit discovery because the unit of analysis matches a model component rather than its activations.

Open questions. (a) Reconstruction error: even good SAEs leave 5–20% of variance unexplained; what is in the residual? (b) Feature universality: are the features Anthropic finds in Claude the same as the features OpenAI finds in GPT-4? (Early universality evidence exists but is fragile.) (c) Feature splitting: how do we choose the granularity? (d) Steering side effects: amplifying a feature changes downstream behavior in ways that often go beyond the feature’s nominal meaning.

37 Attribution Graphs and Circuit-Level Auditing

On the Biology of a Large Language Model (Lindsey et al., March 2025; already cited in Part III) is the most ambitious available demonstration that LLM computation can be made auditable at the circuit level. The companion *Tracing the Thoughts of a Large Language Model* (Lindsey et al., March 2025) presents the same methodology with shorter, more pedagogical case studies. The cluster includes:

- Multi-hop reasoning visible as intermediate activations.
- Forward planning in poetry generation.
- Multilingual concept sharing.
- Documented post-hoc rationalization (the stated chain of thought diverges from the actual mediating activations).

Honest assessment. The methodology is publicly available; the Anthropic team reports satisfying

explanations for about a quarter of prompts. Replication outside Anthropic is in progress. For Noosphere, the practical implications are forward-looking: if attribution graphs become a standard interpretability primitive, the “trace as source of truth” philosophy can extend inside the LLM, not just around it.

38 Influence Functions for LLMs

Roger Grosse et al. (Anthropic 2023, “Studying Large Language Model Generalization with Influence Functions,” arXiv:2308.03296) scaled influence functions — the second-order method for asking “which training examples most affected this output” — to 52B-parameter models. They use EK-FAC (Eigenvalue-corrected Kronecker-Factored Approximate Curvature) to approximate the inverse Hessian-vector products that influence functions require.

Findings.

- Influence patterns are *sparse*: a small fraction of training examples disproportionately drives any given output.
- Influence becomes *more abstract* at scale: smaller models have more keyword-driven influences; larger models have more semantic ones.
- Cross-lingual generalization: training examples in one language influence outputs in another.
- Failure mode: influence decays to near-zero when the order of key phrases is reversed — a striking opacity in what should be order-invariant retrieval.

For Noosphere. If the firm ever wants to ask “which transcripts in our corpus most strongly produced this conclusion,” influence functions are the principled tool. The Anthropic implementation is published; the EK-FAC machinery transfers.

39 The CCS Critique Literature

Volume I cited Burns, Ye, Klein, Steinhardt (2022) on Contrast-Consistent Search. A reader who studies CCS must also study the critiques.

Farquhar et al. (December 2023), “Challenges with unsupervised LLM knowledge discovery,” arXiv:2312.10029. The central finding: existing unsupervised methods on LLM activations do not specifically discover *knowledge*; they discover *whatever feature of the activations is most prominent*. The authors prove theoretically that arbitrary features (not just truth-tracking ones) satisfy CCS’s consistency property, and they demonstrate empirically that CCS makes predictions extremely similar to those of plain PCA. Knowledge discovery in the strong sense remains an open problem.

Implications.

- A direction that satisfies CCS’s logical-consistency constraints is not, by virtue of doing so, a truth direction. It may be a salience direction, a sentiment direction, or a topic direction that happens to be linearly separable.
- For Noosphere, by extension: a learned “contradiction direction” that satisfies geometric desiderata in calibration is not thereby a *semantic* contradiction direction. The QH benchmark’s negative controls are the right tool to defend against this; the firm should publish them.

Follow-up. *Three Concrete Challenges and Two Hopes for the Safety of Unsupervised Elicitation* (Anthropic Alignment, 2026, arXiv:2602.20400) revisits the problem and articulates positive directions under stricter assumptions. Worth reading after Farquhar et al.

40 Function Vectors, Task Vectors, and Activation Steering

A growing family of papers makes operational use of the linear-representation hypothesis by extracting and reusing vectors that encode tasks or behaviors.

Function vectors (Todd et al., 2024, “Function Vectors in Large Language Models,” arXiv:2310.15213): from in-context demonstrations, one can extract a vector at a particular layer/head that, when added back to the activations on a zero-shot input, induces the demonstrated behavior. The vector “packages” the task.

Task vectors via arithmetic (Ilharco et al., 2023, “Editing Models with Task Arithmetic,” arXiv:2212.04089; pre-LLM but the conceptual ancestor): subtract pre-finetune weights from post-finetune weights to get a task vector; arithmetic on these vectors moves models between capabilities.

Activation steering for safety (Rimsky et al., 2024, “Steering Llama 2 via Contrastive Activation Addition,” arXiv:2312.06681; Panickssery et al., 2024): identify a direction from contrastive pairs (refused vs. unrefused, helpful vs. harmful) and add a scaled version of the direction during inference to push the model along that axis.

Persona vectors / model dispositions (Anthropic, ongoing): a research line treating model “character” as a manipulable representation, not just an emergent property of training.

For Noosphere. If contradiction or coherence is geometrically encodable, the function-vector toolkit is the natural way to test it: extract a candidate vector from contrastive examples and verify whether amplifying it induces and suppressing it removes the behavior, at held-out inputs. This is the activation-patching test applied to a candidate Noosphere “coherence direction.” The methodology is well-defined; the firm should adopt it.

41 Knowledge Editing: ROME, MEMIT, and Successors

Meng et al. (*ROME*, 2022; *MEMIT*, 2023, arXiv:2210.07229) introduced rank-one edits to MLP layers that change a model’s factual beliefs (“The Eiffel Tower is in Rome”) with surprising specificity.

Caveats. Subsequent work (Hoelscher-Obermaier et al. 2023, “Detecting Edit Failures in Large Language Models”; Hase et al. 2024) showed that edits often fail to propagate to logically-implied paraphrases, and that the apparent specificity is fragile under distribution shift. Editing is *not* the same as updating, and a firm that uses model editing to correct mistakes should expect leakage and brittleness.

For Noosphere. The firm should not edit its LLM to encode its principles. Principles belong in the symbolic graph and the prompt context, not in the model weights. If it ever did want to edit, it would need to invest in the evaluation methodology that the editing literature has been developing.

42 World Models and the Inner Map

A separate frontier asks whether LLMs learn internal models of the domain they operate over, rather than only surface statistics.

Othello-GPT (Kenneth Li et al., 2023, “Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task,” arXiv:2210.13382): a small transformer trained only on Othello move sequences develops an internal representation of the board state, which linear probes can recover. A clean demonstration that “world models” can emerge from sequence prediction alone.

Chess-GPT and others: subsequent work has extended this to chess and to small language tasks; the pattern — internal latent state recovered by probes — has held up.

For Noosphere. If Claude is doing something like maintaining an internal model of the speakers in a transcript (which speaker holds which view, what has been retracted, what is being entertained hypothetically), the Othello-GPT methodology is the right framework for testing whether the firm’s extractor is reading off a real internal structure or projecting one onto whatever the model produces.

43 Sycophancy, Honesty, and the Disposition Question

Sycophancy (Perez et al., Anthropic 2022, “Discovering Language Model Behaviors with Model-Written Evaluations,” arXiv:2212.09251; Sharma et al., Anthropic 2023, “Towards Understanding Sycophancy in Language Models,” arXiv:2310.13548): models reliably adjust answers to match the apparent views of the questioner. This is a learned RLHF artifact and appears across model families.

Honest prompts and dispositions: a line of work asks whether honesty can be elicited or installed as a stable disposition rather than a moment-to-moment behavior. Park et al. (2023, “AI Deception: A Survey of Examples, Risks, and Potential Solutions,” arXiv:2308.14752); Lin, Hilton, Evans (2022, TruthfulQA).

For Noosphere. A coherence engine whose LLM judge is sycophantic toward the prompt’s evident framing is a coherence engine that scores high whenever the founder agrees with himself. The mitigations are not clever prompting alone; they include adversarial generators (the Volume I local inventory mentions one — it should be central, not auxiliary), red teams, and tests that swap the surface framing of the same proposition to see whether the judgement is stable.

44 Mechanistic Anomaly Detection and ELK

ARC’s *Eliciting Latent Knowledge* (ELK) program (Christiano, Cotra, Xu, 2021, “Eliciting Latent Knowledge,” Google Doc and arXiv) poses the problem of training a reporter that says *what the model believes* rather than *what humans believe* (or what humans would endorse). ELK is the formal articulation of the worry that aligns RLHF can optimize for human-pleasing outputs at the expense of model-honest ones.

The line of work has produced *mechanistic anomaly detection*: detect when a model’s behavior on a deployment input does not match the mechanism that produced its training behavior, even when the output looks normal.

For Noosphere. ELK is far enough upstream that the firm cannot directly use its methods today, but it is the most serious available articulation of why coherence-of-output is not a reliable proxy for

internal model state. Reading the original ELK report is the best single investment to understand what could go wrong with a Noosphere that becomes sufficiently sophisticated to rely on the LLM’s judgement.

45 Recommended reading for Part VII

- Jack Lindsey et al., *On the Biology of a Large Language Model; Tracing the Thoughts of a Large Language Model*. Anthropic, 2025. <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.
- Roger Grosse et al., *Studying Large Language Model Generalization with Influence Functions*. Anthropic 2023. arXiv:2308.03296.
- Sebastian Farquhar et al., *Challenges with unsupervised LLM knowledge discovery*. arXiv:2312.10029, December 2023.
- Eric Todd et al., *Function Vectors in Large Language Models*. ICLR 2024. arXiv:2310.15213.
- Nina Rimsky et al., *Steering Llama 2 via Contrastive Activation Addition*. arXiv:2312.06681, 2024.
- Kevin Meng et al., *Mass-Editing Memory in a Transformer (MEMIT)*. ICLR 2023. arXiv:2210.07229.
- Kenneth Li, Aspen Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, Martin Wattenberg, *Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task*. ICLR 2023. arXiv:2210.13382.
- Ethan Perez et al., *Discovering Language Model Behaviors with Model-Written Evaluations*. ACL 2023. arXiv:2212.09251.
- Mrinank Sharma et al., *Towards Understanding Sycophancy in Language Models*. ICLR 2024. arXiv:2310.13548.
- Peter S. Park et al., *AI Deception: A Survey of Examples, Risks, and Potential Solutions*. Patterns 2024. arXiv:2308.14752.
- Paul Christiano, Ajeya Cotra, Mark Xu, *Eliciting Latent Knowledge*. ARC report, 2021. <https://www.alignment.org/elk/>.
- Anthropic Alignment Team, *Three Concrete Challenges and Two Hopes for the Safety of Unsupervised Elicitation*. arXiv:2602.20400, 2026.

Part VIII — Methodology: How to Test the Hypothesis Rigorously

46 Pre-Registration

Pre-registration is the practice of recording, in advance of seeing the data, what hypotheses you will test, what statistics you will compute, how you will handle missing data, and what would count as confirming or disconfirming evidence. It is the most effective single defense against the researcher degrees of freedom that drive false-positive inflation.

The standard pre-registration template (AsPredicted; Open Science Framework) asks for: hypotheses, dependent variables, conditions, statistical analyses, exclusion criteria, sample size justification, and stopping rule. A pre-registered analysis can still discover surprises; what changes is that exploratory and confirmatory analyses are clearly labelled as such.

For Noosphere. The QH benchmark is a confirmatory enterprise: it is testing whether the contradiction-geometry probe outperforms cosine. The firm should pre-register, for each method version: the dataset, the metric, the threshold (or its calibration procedure), the comparison baselines, the sample size, and the decision rule (“method is adopted if the probe’s balanced accuracy exceeds the cosine baseline by at least 5 percentage points with a 95% CI not crossing zero”). Without this, every benchmark is reportable as either a success or a failure depending on the analysis choice; with it, the firm has an actually testable methodology.

47 Power Analysis with Worked Examples

Section I.3 gave the formula. Here are two worked examples directly applicable to Noosphere.

47.1 Example: contradiction-detector vs. cosine baseline

Suppose the cosine baseline achieves 70% balanced accuracy and you want to detect a 5-point lift to 75%. Plug $p_1 = 0.75$, $p_2 = 0.70$, $\alpha = 0.05$, power 0.80:

$$n \approx \frac{(1.96 \cdot \sqrt{2 \cdot 0.725 \cdot 0.275} + 0.84 \cdot \sqrt{0.75 \cdot 0.25 + 0.70 \cdot 0.30})^2}{0.05^2} \approx 1090.$$

So roughly 1100 examples per arm. For a 3-class problem (cohere / contradict / unrelated), you need this in each of the three pairwise comparisons, or you use a single multinomial test — in either case, several thousand labelled examples.

47.2 Example: Householder ablation

The QH-v1 Householder ablation found that “all variants predicted the same label.” This is a power-zero condition by construction: the variants are not distinguishable. The pre-registration fix is to specify, in advance, *which* variants will be compared, on *which* examples, with *which* threshold-calibration procedure, and *what* difference in calibrated performance would constitute evidence for or against keeping the step.

A defensible ablation protocol:

1. Held-out labelled set of $N \geq 1000$ pairs across at least three domains and two embedding models.

2. Recalibrate the decision threshold on a separate calibration split for each variant; do not reuse the frozen threshold across variants.
3. Report balanced accuracy, AUROC, ECE, and class-conditional accuracy with bootstrap 95% CIs (≥ 1000 resamples).
4. Pre-declare the decision rule (e.g., adopt the simpler variant unless the more complex one improves balanced accuracy by ≥ 0.03 with a 95% CI excluding zero).

48 Proper Scoring Rules and Calibration, Operationalized

Part I.3 gave the definitions. The operational forms for Noosphere:

48.1 Brier and log scores for the coherence engine

If the six-layer engine produces a probability p that two claims cohere, collect (p_i, y_i) pairs from labelled examples and report:

- Mean Brier score $\frac{1}{N} \sum (p_i - y_i)^2$. Lower is better.
- Mean log score $\frac{1}{N} \sum -\log p_i^{y_i} (1 - p_i)^{1-y_i}$. Lower is better. Infinite if $p_i \in \{0, 1\}$ ever disagrees with y_i ; clip to $[\epsilon, 1 - \epsilon]$ in practice.
- Brier skill score relative to a baseline: $1 - \text{Brier}_{\text{method}} / \text{Brier}_{\text{baseline}}$. Positive means improvement.

48.2 Calibration metrics

Report *Expected Calibration Error* (ECE), *Maximum Calibration Error* (MCE), and a reliability diagram (10-bin equal-width, or equal-mass). If raw scores are miscalibrated, apply Platt scaling on a calibration split and report both raw and calibrated metrics.

48.3 Goodhart’s law and the design of un-game-able metrics

Goodhart’s law (originally Charles Goodhart, 1975, in monetary policy; popularized for ML by Manheim & Garrabrant, 2018, “Categorizing Variants of Goodhart’s Law,” arXiv:1803.04585): when a measure becomes a target, it ceases to be a good measure.

Four mechanisms:

1. *Regression Goodhart*: a noisy proxy for a true objective is maximized to a point where the proxy and the true objective diverge.
2. *Extremal Goodhart*: the proxy and the true objective coincide in normal operating ranges but diverge at the extremes.
3. *Causal Goodhart*: intervention on the proxy does not produce the intervention on the true objective that mere correlation would suggest.
4. *Adversarial Goodhart*: optimizers actively seek out proxy-target divergences.

For Noosphere: *coherence score* is a proxy for the true objective (probably “truthfulness of conclusion” or “decision quality”). All four Goodhart mechanisms threaten it. The defenses are not technical but structural: maintain multiple independent metrics, including ones whose manipulation is genuinely

costly (e.g., realized forecast accuracy); periodically audit whether high-coherence outputs are actually better by non-coherence measures; refuse to optimize for coherence in isolation.

49 The Replication Crisis and What to Take From It

John Ioannidis (2005, “Why Most Published Research Findings Are False,” PLoS Medicine 2(8)) argued that under standard scientific practice, more than half of published statistically-significant findings are false positives. The 2015 Open Science Collaboration replication of 100 psychology studies recovered the original effect in roughly 39% of cases; mean effect sizes in replications were about half those originally reported.

The drivers (recapitulating Ioannidis): small samples, low statistical power, many hypotheses tested without correction, researcher degrees of freedom, publication bias, p-hacking, selective reporting.

For Noosphere. The firm builds intellectual capital partly by synthesizing claims from the published literature. *Every claim that rests on a single empirical study with p near 0.05 should be treated as provisional pending replication.* The cascade graph should record, for imported empirical claims, the sample size, the replication status, and any known meta-analytic effect size. Without that, the firm’s principles will inherit the publication-bias-distorted average of the literature, and the geometric coherence engine will efficiently re-package those distortions.

50 Specification Curve and Multiverse Analysis

Specification curve (Simonsohn, Simmons, Nelson, 2020, Nature Human Behaviour): for every reasonable analytic choice (covariates, exclusions, transformations), run the analysis; report the distribution of results across the specification universe. Robustness manifests as a tight cluster of results; fragility manifests as a wide spread.

Multiverse analysis (Steegeen, Tuerlinckx, Gelman, Vanpaemel, 2016): similar in spirit, expanded to data-preparation choices.

For Noosphere. The contradiction-geometry engine has many parameters: which embedding model, which layer (if intra-LLM), which pooling, which normalization, what threshold, how Hoyer sparsity is weighted. A specification-curve presentation of the QH benchmark across these choices would tell you whether the headline result is robust or an artifact of a lucky configuration. This is the methodological lift the firm’s benchmarks most need.

51 Adversarial Collaboration

Adversarial collaboration (Mellers, Hertwig, Kahneman 2001) is the practice of having proponents and skeptics jointly design and run a study whose result they pre-commit to accepting. For methodological claims — and the Quintin Hypothesis is at root a methodological claim — adversarial collaboration is the gold standard. The firm should consider, for any major external assertion about coherence geometry, designing a benchmark with a collaborator known to be skeptical, pre-registering it, and publishing the result.

52 Recommended reading for Part VIII

- Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, David T. Mellor, *The preregistration revolution*, PNAS 2018.
- John P.A. Ioannidis, *Why Most Published Research Findings Are False*, PLoS Medicine 2(8), 2005.
- Open Science Collaboration, *Estimating the reproducibility of psychological science*, Science 349, 2015.
- Uri Simonsohn, Joseph Simmons, Leif Nelson, *Specification curve analysis*, Nature Human Behaviour 4, 2020.
- Sara Steegen, Francis Tuerlinckx, Andrew Gelman, Wolf Vanpaemel, *Increasing transparency through a multiverse analysis*, Perspectives on Psychological Science 11, 2016.
- David Manheim & Scott Garrabrant, *Categorizing Variants of Goodhart's Law*, arXiv:1803.04585, 2018.
- Andrew Gelman & Eric Loken, *The garden of forking paths*, unpublished but widely cited, 2013. <http://www.stat.columbia.edu/~gelman/research/unpublished/forking.pdf>.
- Stuart Ritchie, *Science Fictions: How Fraud, Bias, Negligence, and Hype Undermine the Search for Truth*, Metropolitan Books 2020. The accessible book-length treatment of the replication crisis.
- Daniel Lakens et al., *Justify your alpha*, Nature Human Behaviour 2018.
- Aki Vehtari, Andrew Gelman, Jonah Gabry, *Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC*, Statistics and Computing 27, 2017. arXiv:1507.04544.

Part IX — Engineering and Tooling

This part is short and practical. The references are documentation rather than papers; the goal is to point at the tools an engineer maintaining Noosphere should know.

53 The Embedding Landscape

53.1 API embedding providers (production)

OpenAI `text-embedding-3-large` (3072d) and `text-embedding-3-small` (1536d). Matryoshka representation learning (Kusupati et al., 2022, arXiv:2205.13147) means you can truncate to any prefix length with graceful degradation.

Voyage AI `voyage-3-large` and the `voyage-3-lite/voyage-large-2-instruct` family. As of mid-2025 near or at the top of the MTEB English leaderboard; particularly strong on retrieval.

Cohere `embed-english-v3.0` and the multilingual `embed-multilingual-v3.0`.

Anthropic does not offer first-party embeddings as of mid-2026; Noosphere using Claude for extraction and a separate provider for embeddings is the standard pattern.

53.2 Open-source embeddings

BGE / FlagEmbedding (BAAI): `bge-large-en-v1.5`, `bge-m3` (multilingual + multi-vector + sparse). Among the strongest open-source options.

E5 / Multilingual-E5 (Microsoft).

GTE (Alibaba).

Nomic `embed-text-v1.5` (Nomic): open weights and open training data.

Stella, `jina-embeddings-v3`, `mxbai-embed-large-v1`: actively maintained competitive open models.

53.3 MTEB: the benchmark

The *Massive Text Embedding Benchmark* (Muennighoff, Tazi, Magne, Reimers, 2022, arXiv:2210.07316; ongoing leaderboard at HuggingFace). 56+ tasks spanning retrieval, classification, clustering, STS, summarization, pair classification, reranking, and bitext mining. The right place to start when choosing an embedding model. Read the per-task subscores, not just the average: an embedding model that excels at retrieval but underperforms on STS will give different downstream results.

53.4 Recommended reading and tools

- Niklas Muennighoff, Nouamane Tazi, Loïc Magne, Nils Reimers, *MTEB: Massive Text Embedding Benchmark*. arXiv:2210.07316. <https://huggingface.co/spaces/mteb/leaderboard>.
- Aditya Kusupati et al., *Matryoshka Representation Learning*, NeurIPS 2022. arXiv:2205.13147.
- Anthropic, *sentence-transformers* package documentation, <https://www.sbert.net/>.

54 Vector Databases

FAISS (Meta): library, not a database. Industry standard for in-process nearest neighbor on indexed vectors. Supports IVF, HNSW, PQ, OPQ. Free, GPU and CPU.

HNSW (Malkov & Yashunin, 2016, “Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs,” arXiv:1603.09320): the algorithm, not a product; implemented in FAISS, hnsplib, pgvector, Pinecone, ChromaDB.

ChromaDB: open-source, embedded by default, simple API. The right starting point for a small-team prototype.

pgvector: PostgreSQL extension. Lets you keep claims, embeddings, and metadata in one transactional store. The right starting point if relational schema discipline matters.

Qdrant, Weaviate, Milvus: open-source distributed vector databases. Used when the corpus grows past the in-process scale of ChromaDB/pgvector.

Pinecone, Cloudflare Vectorize: managed services. Pay-for-scale without ops overhead.

For Noosphere at firm scale, pgvector or ChromaDB is more than adequate; the relevant engineering question is not throughput but *schema*: which fields besides the vector itself (model name, model version, embedding pipeline version, text hash, source claim id, embedding date) you store alongside.

55 Hybrid Retrieval

Pure dense retrieval (cosine over embeddings) misses lexically rare terms; pure sparse retrieval (BM25) misses semantic paraphrases. Hybrid combines them:

Reciprocal Rank Fusion (Cormack, Clarke, Büttcher, 2009): combine ranked lists by $\sum_i 1/(k + \text{rank}_i)$. Simple, surprisingly effective.

ColBERT / ColBERTv2 (Khattab, Zaharia, 2020; Santhanam et al. 2022, arXiv:2112.01488): late-interaction dense retrieval; per-token embeddings with MaxSim aggregation. Better quality than pooled-vector retrieval at higher compute cost.

SPLADE (Formal, Lassance, Piwowarski, Clinchant, 2021, arXiv:2107.05720): sparse-lexical embeddings learned with masked language modeling. Combines the recall properties of BM25 with learned semantics.

56 Knowledge-Graph Stores

Neo4j: the dominant property-graph database. Cypher query language. Good UI for inspection; less natural for OWL/RDF if the ontology becomes formal.

TigerGraph, ArangoDB: alternatives at scale.

RDF triple stores: GraphDB (Ontotext), Blazegraph (now defunct, but Wikidata’s historical store), Apache Jena, Virtuoso. Choose these if you commit to RDF/OWL.

NetworkX: a Python library, not a database. Right for prototyping and small graphs; not for persistent or concurrent operation.

For Noosphere at current scale, NetworkX-in-process with a serialized checkpoint on disk is adequate, but the firm should know what it would take to migrate to Neo4j or a triple store if the graph grows

or if multi-user access becomes important.

57 Interpretability Tooling

TransformerLens (Nanda et al. 2022, open source): the dominant library for mechanistic interpretability research. Provides hooks for intercepting and modifying activations at every component.

nnsight (Bau et al. 2024, NDIF, arXiv:2407.14561): scales the TransformerLens API to large models with remote-execution; the right tool for working with frontier-scale open models.

SAELens: the canonical library for training and applying sparse autoencoders on transformer activations.

baukit, **captum**, **pyvene**: alternative libraries with different abstractions.

58 Evaluation and NLI Models

DeBERTa-v3-large fine-tuned for NLI on combined MNLI / FEVER / ANLI / DocNLI: the dominant open NLI model in 2024–2026.

nli-deberta-v3-large (cross-encoder/nli-deberta-v3-large on HuggingFace) is what most production coherence pipelines use as the NLI layer.

UniEval (Zhong et al., 2022, arXiv:2210.07197): multi-dimensional LLM-judge for evaluation (coherence, consistency, fluency, relevance).

For Noosphere. The first layer of the six-layer coherence engine should specify which NLI model it uses; this is a versioned choice. DeBERTa-v3-NLI is the default; if the firm ever moves to a stronger NLI model, the QH benchmark needs to be re-run.

59 Production Patterns and Their Names

RAG (Retrieval-Augmented Generation): retrieve relevant documents, condition generation on them. Lewis et al. 2020. Now standard.

Self-RAG, **CRAG**, **GraphRAG**: variants that add self-critique (Asai et al. 2024), correction (Yan et al. 2024), and graph traversal (Microsoft GraphRAG, Edge et al. 2024, arXiv:2404.16130) on top of plain RAG.

Function calling / tool use: the model emits structured calls to external functions, whose results are fed back into the context. The mechanism by which Noosphere could let the LLM query its own knowledge graph or vector store mid-generation.

Caching layers: prompt caching (Anthropic), KV-cache reuse, and embedding caches dramatically reduce cost and latency in production.

Tracing and observability: LangSmith, Helicone, Phoenix, custom OpenTelemetry instrumentation. Whatever the firm uses must capture the fields enumerated in Volume I’s Section 20.1.

60 Recommended reading and documentation for Part IX

- *HuggingFace transformers documentation*, ongoing. <https://huggingface.co/docs/transformers/index>.
- *sentence-transformers documentation*, ongoing. <https://www.sbert.net/>.
- Yury Malkov & Dmitry Yashunin, *Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs*. arXiv:1603.09320, 2016/2018.
- Omar Khattab & Matei Zaharia, *ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT*. SIGIR 2020. arXiv:2004.12832.
- Keshav Santhanam et al., *ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction*. NAACL 2022. arXiv:2112.01488.
- Darren Edge et al., *From Local to Global: A Graph RAG Approach to Query-Focused Summarization*. Microsoft 2024. arXiv:2404.16130.
- Neel Nanda et al., *TransformerLens documentation*. <https://transformerlensorg.github.io/TransformerLens/>.
- Jaden Fiotto-Kaufman et al., *NNsight and NDIF: Democratizing Access to Foundation Model Internals*. arXiv:2407.14561, 2024.

Part X — Expanded Bibliography

This bibliography is a second tier beyond Volume I’s source inventory: items that did not appear in Volume I but that the reader who has worked through Volume II should know exist. Organized by section. Where a freely available PDF exists, the URL is given; otherwise the citation is sufficient to find the work.

For Part I (Mathematics)

- Sanjeev Arora, Elad Hazan, Satyen Kale, *The Multiplicative Weights Update Method: A Meta-Algorithm and Applications*, Theory of Computing 8, 2012. The single most useful unification of gradient-based and information-theoretic algorithms.
- Jean-Pierre Aubin, *Optima and Equilibria: An Introduction to Nonlinear Analysis*, Springer 1998. A more general lens on optimization.
- Michael I. Jordan, *An Introduction to Variational Methods for Graphical Models*, Machine Learning 37, 1999. The bridge from probabilistic graphical models to modern variational inference.
- Sham Kakade, *A Natural Policy Gradient*, NeurIPS 2001. The information-geometric formulation of RL gradient methods.
- Martin J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge 2019. The technical reference for high-dim concentration.

For Part II (LLM Internals)

- Tom Brown et al., *Language Models are Few-Shot Learners* (GPT-3). NeurIPS 2020. arXiv:2005.14165.
- OpenAI, *GPT-4 Technical Report*, 2023. arXiv:2303.08774.
- Hugo Touvron et al., *Llama 2: Open Foundation and Fine-Tuned Chat Models*, 2023. arXiv:2307.09288.
- Albert Q. Jiang et al., *Mistral 7B*, 2023. arXiv:2310.06825.
- Mosaic ML & others, *MPT and follow-ups*; see ongoing reports.
- Aakanksha Chowdhery et al., *PaLM: Scaling Language Modeling with Pathways*, JMLR 2023. arXiv:2204.02311.
- Noam Shazeer et al., *Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer*, ICLR 2017. arXiv:1701.06538. The ancestor of modern MoE.
- Mostafa Dehghani et al., *Universal Transformers*, ICLR 2019. arXiv:1807.03819.
- Tri Dao et al., *FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness*, NeurIPS 2022. arXiv:2205.14135. The reason production transformers can be fast and exact at once.
- Albert Gu & Tri Dao, *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*, 2023. arXiv:2312.00752. The leading non-transformer architecture.
- Albert Gu et al., *Efficiently Modeling Long Sequences with Structured State Spaces (S4)*. ICLR 2022. arXiv:2111.00396.

For Part III (Embedding Geometry)

- Tianlin Liu et al., *Same Pre-training Loss, Better Downstream: Implicit Bias Matters for Language Models*, ICML 2023. The bias-variance view of pretraining loss vs. downstream behavior.
- Aaditya Ramdas, Aaditya, *Foundations of Causal Inference* lecture notes (CMU), free online.
- Mostafa Dehghani et al., *The Efficiency Misnomer*, ICLR 2022.
- Bowen Cui et al., *Survey on Sparse Autoencoders for LLMs*, arXiv 2025 onwards (a rapidly growing area; pin a recent survey).
- Joseph Bloom, ongoing posts at LessWrong/AlignmentForum on SAE training and feature analysis. Less formal but the closest thing to a working laboratory notebook in the field.

For Part IV (NLP Extraction)

- Christopher Manning et al., *The Stanford CoreNLP Natural Language Processing Toolkit*, ACL 2014 demo paper. The system, but more importantly a canonical reference for the classical pipeline architecture.
- Yoav Goldberg, *Neural Network Methods for Natural Language Processing*, Morgan & Claypool 2017. The cleanest mid-decade textbook on the transition from classical to neural NLP.
- Iz Beltagy et al., *Longformer: The Long-Document Transformer*, 2020. arXiv:2004.05150. The long-context transformer family.
- Tianyi Zhang et al., *BERTScore: Evaluating Text Generation with BERT*, ICLR 2020. arXiv:1904.09675.
- William Lewis et al., *The Anatomy of FEVER*, 2018. The fact verification benchmark.
- Ofir Press et al., *Measuring and Narrowing the Compositionality Gap in Language Models*, EMNLP Findings 2023. arXiv:2210.03350.

For Part V (Graphs, Ontologies, Formal Logic)

- Daniel Lehmann, *Belief Revision, Revised*, 1995. The early critique and refinement of AGM.
- Henry Prakken, *An Abstract Framework for Argumentation with Structured Arguments*, Argument & Computation 1, 2010. The ASPIC⁺ foundation.
- Phan Minh Dung et al., *Assumption-based argumentation*, multiple papers from the 2000s onward.
- Pierre Marquis, Odile Papini, Henri Prade (eds.), *A Guided Tour of Artificial Intelligence Research, Volume 1: Knowledge Representation, Reasoning and Learning*, Springer 2020. The contemporary reference handbook.
- Bart Selman et al., *Local Search Strategies for Satisfiability Testing*, 1993. The ancestor of modern SAT solvers, which the firm will need if its ontology grows.
- Diego Calvanese et al., *Ontology-Based Data Access: A Survey*, IJCAI 2018. The bridge from formal ontology to database-style querying.
- Pedro Domingos & Daniel Lowd, *Markov Logic: An Interface Layer for Artificial Intelligence*, Morgan & Claypool 2009. Book-length treatment.

For Part VI (Epistemology)

- Keith Lehrer, *Theory of Knowledge*, 2nd ed., Westview 2000. The systematic statement of Lehrer’s coherentism.
- Laurence Bonjour, *Knowledge of the External World*, Cambridge 2003. Bonjour’s later, post-coherentist position.
- Ernest Sosa, *Knowing Full Well*, Princeton 2011. The unified statement of virtue-theoretic reliabilism.
- Timothy Williamson, *Knowledge and Its Limits*, Oxford 2000. The most influential contemporary epistemology monograph; knowledge-first rather than belief-first.
- Hilary Putnam, *The Many Faces of Realism*, Open Court 1987. Pragmatic-internal-realist alternative to correspondence.
- Edward Craig, *Knowledge and the State of Nature*, Oxford 1990. The genealogical/functionalist account of why we have a concept of knowledge in the first place.
- David Christensen, *Putting Logic in Its Place: Formal Constraints on Rational Belief*, Oxford 2004. On the conditions under which logic constrains coherent credences.
- Branden Fitelson, *Studies in Bayesian Confirmation Theory*, PhD dissertation 2001. The technical foundation of contemporary confirmation theory.

For Part VII (Frontier)

- Anthropic’s *Transformer Circuits* thread (ongoing), <https://transformer-circuits.pub>. Pin the index page; new entries roughly monthly.
- DeepMind’s interpretability publications (Senthooan Rajamanoharan, Anthropic + DeepMind crossover, ongoing).
- Apollo Research publications: alignment-focused interpretability.
- Redwood Research publications: causal scrubbing methodology.
- EleutherAI’s interpretability papers; the openness counterpart to the closed-lab agenda.
- Buck Shlegeris, Ryan Greenblatt et al., AI Control series, arXiv 2023–2024.
- Yoshua Bengio et al., *International AI Safety Report*, 2025. Multi-author state-of-the-field document.

For Part VIII (Methodology)

- Andrew Gelman & Jennifer Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models*, Cambridge 2007. The applied statistics reference.
- Larry Wasserman, *All of Statistics: A Concise Course in Statistical Inference*, Springer 2004. Short and dense; the right reference for a non-statistician who needs everything.
- Aleks Jakulin & Aki Vehtari, ongoing posts on Stan / Bayesian workflows.
- Donald T. Campbell & Julian C. Stanley, *Experimental and Quasi-Experimental Designs for*

Research, Houghton Mifflin 1963. The classical reference on internal and external validity.

- Daniel Kahneman, Olivier Sibony, Cass Sunstein, *Noise: A Flaw in Human Judgment*, Little Brown 2021. Specifically on the gap between calibration and noise reduction.

Bridges and meta-references

- Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson 2020. The encyclopedic textbook; useful as a sanity check that the firm has not invented a wheel.
- Christopher Bishop, *Pattern Recognition and Machine Learning*, Springer 2006. The classical ML reference.
- Kevin P. Murphy, *Probabilistic Machine Learning: An Introduction* (2022) and *Probabilistic Machine Learning: Advanced Topics* (2023), MIT Press. Free draft PDFs at the author's site. The modern Bishop.
- Aileen Nielsen, *Practical Fairness*, O'Reilly 2020. The operational counterpart to the methodology literature.
- Cathy O'Neil, *Weapons of Math Destruction*, Crown 2016. The case for why metric design is an ethical decision.

Closing: What Volume II Has Tried to Do

Volume I gave the conceptual map of Noosphere’s substrates. Volume II has tried to expand that map in three directions.

First, the math. The Householder reflection becomes a meaningful object only when you can derive it from projection, see why it is the building block of QR factorization, and recognize that its operational use in Noosphere is a specific bet about the geometric structure of contradiction. Cosine similarity becomes a meaningful object only when you understand that it is the inner product of normalized vectors and that, in high dimensions, it has a particular noise distribution that any threshold must respect. Probability becomes a meaningful object only when you separate “forecast,” “calibrated probability,” and “decision-relevant expectation,” and when you can write down a power analysis.

Second, the frontier. Sparse autoencoders, attribution graphs, influence functions, function vectors, and the critique line on Contrast-Consistent Search are not optional reading for a firm whose entire premise is that LLM representations carry geometric structure that can be exploited. *Toward Monosemanticity*, *Scaling Monosemanticity*, and *On the Biology of a Large Language Model* are, between them, the strongest positive case that LLMs encode interpretable structure. *Challenges with Unsupervised LLM Knowledge Discovery* is the strongest negative case that current methods do not yet identify that structure reliably. A serious Quintin-Hypothesis programme must engage all of them.

Third, the methodological apparatus. The QH-v1 zero-power ablation is a concrete instance of a general problem: when researcher degrees of freedom exceed evidential constraints, the resulting findings can mean almost anything. Pre-registration, proper scoring rules, calibrated probabilities, specification curves, and adversarial collaboration are the disciplines that turn a coherence engine into an experimental science. They are also the disciplines that distinguish a methodology from a marketing claim.

A reader who has internalized Volume I and worked through Volume II should now be able to (a) read any frontier paper in this list without major gaps, (b) design controlled experiments to test specific Noosphere hypotheses, (c) recognize when a result is real and when it is an artifact of dimensionality, calibration, or model priors, and (d) hold the firm’s own methodology to the standards the firm holds the published literature to. The Quintin Hypothesis is sharp enough to be wounded by the right experiments. The contribution of Volume II is the apparatus to design those experiments.

The reading frontier moves; reorientation is constant. The most reliable single practice for keeping current is to subscribe (in a curated way) to: `transformer-circuits.pub` (Anthropic interpretability), the `cs.CL` and `cs.LG` arXiv categories filtered by venue or author, the ACL Anthology for NLP, the AAAI/IJCAI/KR proceedings for knowledge representation, and the Philosophical Studies / Synthese journals for the analytic-epistemology component. The methods will change in detail; the mathematical and methodological apparatus assembled here should remain stable for at least another half-decade.